



RESEARCH

On optimal upper bound for the settling time of fixed-time stable systems and its application in secure communication

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Abstract This paper proposes a Lyapunov-based fixed-time stability (FXTS) theorem for nonlinear dynamical systems. An upper bound estimate of the settling time and the sufficient condition for FXTS of nonlinear systems are provided. An optimality criterion is derived to obtain the minimum value of the settling time's upper bound estimate. Meanwhile, an insight into accurate estimation of the upper bound of settling time is provided. Based on the FXTS theorem, non-singular terminal sliding surfaces (NSTSM) and corresponding controllers are constructed to make the trajectories of the perturbed nonlinear dynamical system reach the equilibrium point robustly in some fixed-time. A secure communication scheme based on fixed-time synchronization of chaotic systems is also proposed as an application of the theoretical results obtained in this study. Numerical simulations are carried out for the illustration and validation of theoretical claims.

Keywords Fixed-time stability · Sliding mode control · Optimality · Synchronization · Secure communication

1 Introduction

Recently, finite-time stability (FTS) has gained significant attention from researchers in nonlinear control systems. FTS, introduced by Bhat and Bernstein [1], offers the advantage that it guarantees the state trajectories of a system to converge to the desired state within a finite time, which can be particularly useful in real-world control problems such as robot manipulators [2], networked control systems [3], underactuated unmanned underwater vehicles (UUVs) [4], power systems [5], quad-rotor UAVs [6], and many others. However, FTS does have some limitations. One limitation is that it depends on the initial conditions and thus requires complete information about the system's initial conditions. This means that the convergence or settling time may vary for different initial conditions. Also, in many real-world systems, obtaining accurate measurements of all initial states may be challenging or impossible, leading to poor estimates of the settling time and thus reducing its practical implementation.

To overcome these sorts of problems, the term fixed-time stability (FXTS) has been coined by Polyakov [7], where a new form of Lyapunov inequality $\dot{V} \leq -b_0 (V(x))^\alpha - c_0 (V(x))^\beta$ was considered with positive real constants $b_0, c_0, \alpha \in (0, 1)$ and $\beta > 1$. If the system is globally FTS and the settling time function is bounded and constant regardless of any initial conditions, then the system is called FXTS. The independence of settling time on initial conditions in the

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concept of FXTS has attracted a lot of researchers and has been implemented more practically and accurately, such as in switched systems [8], impulsive systems [9], smart grids [10], multiagent systems [11], secure communication [12], spacecraft attitude control [13] and so on. Researchers usually focus on two aspects of FXTS in their studies: (i) the sufficient condition for FXTS and (ii) the upper limit estimate formula for the settling time. Subsequently, after the work of Polyakov to give a sufficient condition as well as the estimate of the upper bound of settling time, several works have been reported on FXTS. Parsegov et al. [14] proposed an FXTS criterion, which could be deemed a particular case of the one from [7]. The upper bound estimate for the settling time they derived was smaller than that of [7]. In [15], a new sufficient condition for FXTS $\dot{V} \leq -b_0 (V(x))^\alpha - c_0 (V(x))^\beta - a_0 V(x)$ with positive real constants $a_0, b_0, c_0, \alpha \in (0, 1)$ and $\beta > 1$ is provided. Due to the presence of the negative linear term, it gives a faster convergence rate than that of [7]. Further, [16] provides a more conservative estimate for the upper bound of settling time in comparison to that of [15]. In [17], another form of FXTS $\dot{V} \leq a_0 V(x) - b_0 (V(x))^\alpha - c_0 (V(x))^{\beta + \text{sign}(V(x)-1)}$, with real constants $a_0, b_0, c_0, \alpha, \beta$ satisfying $b_0, c_0 > 0, \alpha \in (0, 1)$ and $\beta > 1$ is provided using the signum function. If we consider $a_0 < 0$, [17] will give a faster rate of convergence than that of [15] due to the presence of the signum function. Furthermore, some other interesting results, discussions, and applications on FXTS can be found in [18, 19].

On the other hand, since one can convert the synchronization problem of nonlinear systems into the stability problem of the error system of the drive and response systems, the synchronization of dynamical systems is an excellent application for the results on stability. In recent years, there have been some results about the fixed-time synchronization of dynamical systems [17, 20, 21]. In [20], a new sufficient condition, along with the upper bound of settling time, has been applied to synchronize the memristive neural network in fixed time. In [21], fixed-time synchronization of a complex network with impulsive effects is considered.

Moreover, to control the system trajectories according to our needs, some inputs, known as the controller, must be implemented. Accordingly, different kinds of controllers such as active control [22], backstepping control [23], adaptive control [24], saturation control

[25], PID control [26], sliding mode control [27, 28] have come into existence. However, because of the versatility of sliding-mode control (SMC), it has been widely used in numerous uncertain systems, mainly because of its advantages, such as robustness, order reduction, and simplicity in design and implementation. The first step in performing SMC design is to define a switching surface that specifies the desired convergence property. The second step is to design a sliding-mode controller to drive the system states to the selected manifold, which is unaffected by uncertainties or disturbances [29, 30]. In addition, terminal sliding mode control (TSMC) was developed by [31, 32] to achieve finite-time convergence of the system dynamics on sliding surfaces to avoid the infinite convergence time of conventional SMCs. Although finite-time convergence in TSMC is superior to conventional SMC, it has the disadvantage that TSMC may not provide the same convergence performance when the system states are far away from equilibrium. [33] have adopted the fast terminal sliding mode (FTSMC) concept to address this issue, which guarantees quick transient convergence far and near the equilibrium. Although the TSMC/FTSMC with the property of finite or fixed-time convergence has many benefits, it has an inherent singularity problem because of the non-differentiability of the conventional finite or fixed-time sliding variables. To overcome this drawback, researchers have made several efforts in recent times to construct non-singular terminal sliding surfaces [34–37]. The papers referenced [34–37] propose different novel approaches for stabilizing uncertain systems using non singular terminal sliding mode (NSTSM) to achieve FXTS, which means that the system trajectories reach a stable state within a pre-specified time, regardless of initial conditions. Articles [34, 35] propose new NSTSM for second-order uncertain systems, and the papers [36, 37] provide novel designing of terminal non-singular sliding mode surfaces to increase the convergence rate of the trajectories.

Further, Chaos can be described as a surprising but delightful nonlinear aspect to come across constantly during the day. Two of the most important fundamental properties of Chaos are (i) its sensitive dependence on the initial condition, which elaborately describes that the trajectories with close-by initial conditions move apart rapidly and (ii) dense periodic points [38]. It is worth mentioning that the proposal of the OGY approach by Ott et al. [39] in 1990 was the first to think

of controlling chaotic systems. After that, enormous efforts are made to stabilize chaotic systems. To cope with the vast number of applications of chaotic systems, researchers have proposed several new chaotic systems with more complexity in the recent past [40,41]. Among several applications of Chaos, such as in living systems [42], optical cryptography [43], and secure communication [44], the field of secure communication via chaotic synchronization has grown by leaps and bounds in the recent past. The principle is that if we have two chaotic systems, where one of the variables from the first system enslaves the second, this chaotic variable works as a carrier for a message. We can decrypt the encrypted signal by synchronizing the enslaver and enslaved systems. In addition, there are three fundamental techniques. (1) Chaos masking (CMA) [45]; (2) Chaos modulation (CMO) [46]; and (3) Chaos shift keying (CSK) [47]. In the CMA scheme, messages are added to the Chaos generated by the chaotic source; thus, the messages do not modify the Chaos, whereas, in the CMO scheme, the message is added to the Chaos-generating system and thus modifies the Chaos. Further, the CSK scheme modifies a Chaos generator parameter by alternating between ones and zeros. In the present article, we have focused on the Chaos masking scheme, considering a newly defined more complex chaotic system defined in [41] and presented its simple implementation and high security.

Based on the above discussion, authors found one notable aspect of most of the articles on FXTS, such as [7,14,16,17,35], is that there are no dependencies among the parameters. Thus, they are chosen arbitrarily, which may reduce the practical applicability of the proposed approaches. For example, the actual convergence time of the corresponding system trajectories is too less than that of the theoretical upper bound. To overcome this drawback to a certain extent, we have provided a sufficient condition, by introducing a new parameter ϵ , for FXTS, $\dot{V} \leq a_0 V(x) - b_0 (V(x))^\alpha - c_0 (V(x))^{\beta + \text{sign}(V(x) - \epsilon)}$, where $a_0, b_0, c_0, \alpha, \beta, \epsilon$ are real constants satisfying $0 < \alpha < 1, \beta > 1, b_0, c_0, \epsilon > 0, a_0 < \min \{b_0, \frac{c_0}{2}, c_0 \epsilon^\beta, b_0 \epsilon^{-1}\}$ and particular values of ϵ are provided depending on the other parameters. This will lead to more effective stabilization of uncertain systems within a fixed time and potentially improve the practical performance of the proposed methods. Thus, being motivated by the above, the main contributions of the present article can be concluded as follows:

1. A Lyapunov-based sufficient condition for FXTS of a nonlinear dynamical system is provided, along with an estimate of the upper bound of settling time. Further, an optimal condition on one of the parameters ϵ is provided to find the minimum value of the upper bound estimate of settling time. Thus, a strict equality relation on ϵ is given in terms of the other parameters.
2. Moreover, the difference between the true convergence time and the upper bound estimate of the settling time for the optimal case is smaller than that for $\epsilon = 1$, which is studied in the article [17]. In other words, one can say that an accurate estimation of settling time is obtained here in comparison to the article [17].
3. Based on the derived Lyapunov form, non-singular terminal sliding surfaces and corresponding robust controllers concerning bounded disturbances are provided. A secure communication scheme based on the fixed-time synchronization of chaotic systems has been put forward for the application of the present theoretical results.
4. The analysis of the proposed sliding surface is done corresponding to its different sets of parameters. In addition, its effectiveness has been demonstrated by comparison with some recently developed sliding surfaces. Apart from these, various simulated results have illustrated other theoretical results.

The remaining portion of the article is organized as follows. After some brief discussion on preliminaries in Sect. 2, the main results are provided in Sect. 3. Section 4 is dedicated to the designing of the sliding surfaces and controllers. In Sect. 5, the application part of the results in Chaos masking is provided, whereas Sect. 6 consists of numerical simulations to justify the truthfulness of the theoretical claims. Section 7 highlights some concluding remarks.

2 Preliminaries

Notations: Throughout the article we denote, $\mathbb{R}$ as the set of real numbers; $\mathbb{R}_+$ as the set of non-negative real numbers and $\mathbb{R}^n$ is the n -dimensional real number space equipped with the Euclidean norm $|\cdot|$. Also, for $x \in \mathbb{R}$, $\text{sign}(x)$ denotes the signum function. It is defined as:

$$\text{sign}(x) = \begin{cases} -1, & x < 0 \\ [-1, 1], & x = 0 \\ 1, & x > 0 \end{cases}$$

Consider the autonomous system:

$$\dot{x} = F(x), x(0) = x_0 \quad (1)$$

where $x \in \mathbb{R}^n$ is the system states and $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a non-linear function which determines the dynamics of the system. Moreover, $F(x)$ can be discontinuous, and the existence and uniqueness of the solution in forward time is understood in the sense of Filippov [48]. Further, without any loss of generality, the origin $x = 0$ is considered as the equilibrium point of (1).

All the stability notations defined and referred to hereafter are global.

Definition 1 (Stability Notations) [7]: The origin of system (1) is said to be,

- **Finite-time stable:** if it is Lyapunov stable and for any $x_0 \in \mathbb{R}^n$ there exist $0 \leq \tau(x_0) < \infty$ such that the solution $x(t, x_0)$ satisfies $x(t, x_0) = 0$ for all $t \geq \tau(x_0)$. Further, the function $T(x_0) = \inf \{ \tau : x(t, x_0) = 0, \forall t \geq \tau(x_0) \}$ is known as the *settling time function*.
- **Fixed-time stable:** if the settling time function $T(x_0)$ is bounded on $\mathbb{R}^n$ along with the Finite-time stability of the origin, i.e., $\exists T_{fix} > 0$ such that $T(x_0) \leq T_{fix} < \infty$ for all $x_0 \in \mathbb{R}^n$.

Lemma 1 [49] *If one have $x_l \geq 0$, ($l=1, 2, \dots, n$), $\alpha \in (0, 1)$ and $\beta > 1$, then it can be concluded that:*

$$\begin{aligned} \sum_{l=1}^n x_l^\alpha &\geq \left(\sum_{l=1}^n x_l \right)^\alpha \text{ and } \sum_{l=1}^n x_l^\beta \\ &\geq n^{1-\beta} \left(\sum_{l=1}^n x_l \right)^\beta. \end{aligned}$$

Lemma 2 [1] *Let $V(x)$ be a regular, positive definite and radially unbounded function which satisfies the following differential inequality:*

$$\frac{d}{dt} V(x) \leq -b_0 (V(x))^\alpha,$$

for positive real constants b_0 and $\alpha \in (0, 1)$. Then, the origin of system (1) is finite time stable and the settling time function satisfies

$$T(x_0) \leq \frac{(V(x_0))^{1-\alpha}}{b_0(1-\alpha)}.$$

3 Main results

This section of the manuscript consists of the main theoretical findings of the article. More specifically, at first, a Lyapunov-inequality is proposed for fixed-time stability of the system (1). Then an optimal upper bound of the settling time is provided, followed by some insight into an accurate estimation of the settling time in comparison to the article [17].

3.1 Fixed-time stability

In this subsection of the article, a Lyapunov sufficient condition is provided for the fixed-time stability of the system (1), along with some notes pointing to its efficiency.

Theorem 1 *Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be a regular, positive definite and radially unbounded function whose time derivative along the solution trajectories of system (1) satisfies:*

$$\begin{aligned} \frac{d}{dt} V(x) &\leq a_0 V(x) - b_0 (V(x))^\alpha \\ &\quad - c_0 (V(x))^{\beta + \text{sign}(V(x) - \epsilon)} \end{aligned} \quad (2)$$

for some real constants $\alpha, \beta, \epsilon, a_0, b_0, c_0$ satisfying $0 < \alpha < 1, \beta > 1, b_0, c_0, \epsilon > 0$ and $a_0 < \min \{ b_0, \frac{c_0}{2}, c_0 \epsilon^\beta, b_0 \epsilon^{-1} \}$, then the origin of system (1) is fixed-time stable and the estimated upper bound of settling time function is

$$\begin{aligned} T(x_0) &\leq T_{fix} \\ &= \begin{cases} \frac{1}{a_0(1-\alpha)} \ln \left(\frac{b_0}{b_0 - a_0 \epsilon^{1-\alpha}} \right) + \frac{\epsilon^{-\beta}}{\beta c_0}, & a_0 < 0 \\ \frac{1}{b_0(1-\alpha)} \epsilon^{1-\alpha} + \frac{\epsilon^{-\beta}}{\beta c_0}, & a_0 = 0 \\ \frac{1}{a_0(1-\alpha)} \ln \left(\frac{b_0}{b_0 - a_0 \epsilon^{1-\alpha}} \right) + \frac{1}{\beta(c_0 \epsilon^\beta - a_0)}, & a_0 > 0 \end{cases} \end{aligned} \quad (3)$$

Proof The proof is divided into two parts. In the first part we have shown the FTS of the system (1) and in the second part, the estimate of the settling time has been given. For simplicity of computation, let us assume $V = V(x)$.

Now, if $a_0 \leq 0$, we have

$$\frac{dV}{dt} \leq -b_0 V^\alpha - c_0 V^{\beta + \text{sign}(V - \epsilon)} = -b_0 V^\alpha < 0.$$

If $a_0 > 0$, the discussion breaks down into following parts

When $0 < V < \min\{\epsilon, 1\}$, we have

$$\frac{dV}{dt} \leq a_0 V - b_0 V^\alpha < -(b_0 - a_0) V^\alpha < 0. \quad (4)$$

When $V > \max\{\epsilon, 1\}$, one gets

$$\frac{dV}{dt} \leq a_0 V - c_0 V^{\beta+1} < -(c_0 - a_0) V^{\beta+1} < 0. \quad (5)$$

When $\min\{\epsilon, 1\} \leq V \leq \max\{\epsilon, 1\}$, it can further be subdivided into two parts as follows:

If $\epsilon < 1$,

$$\frac{dV}{dt} \leq a_0 V - b_0 V^\alpha - c_0 V^{\beta-1} < -(b_0 - a_0) V^\alpha < 0.$$

If $\epsilon > 1$,

$$\begin{aligned} \frac{dV}{dt} &\leq a_0 V - b_0 V^\alpha - c_0 V^{\beta-1} < a_0 \epsilon - b_0 - c_0 V^{\beta-1} \\ &< -c_0 V^{\beta-1} < 0. \end{aligned} \quad (6)$$

It follows from the above discussion that $\dot{V} < 0$ and from Lemma 2, the origin of system (1) is finite-time stable for any real values of a_0 . In order to find the settling time of the system (1), we consider the following two cases:

Case 1. When $0 < V(x_0) \leq \epsilon$: Here, we get

$$\begin{aligned} T(x_0) &\leq \int_0^{V(x_0)} \frac{dV}{-a_0 V + b_0 V^\alpha + c_0 V^{\beta-1}} \\ &\leq \int_0^\epsilon \frac{dV}{-a_0 V + b_0 V^\alpha + c_0 V^{\beta-1}} \\ &\leq \frac{1}{(1-\alpha)} \int_0^{\epsilon^{1-\alpha}} \frac{dV}{-a_0 V + b_0}. \end{aligned} \quad (7)$$

For $a_0 = 0$, one can get from Eq. (7)

$$T(x_0) \leq \frac{1}{b_0(1-\alpha)} \epsilon^{1-\alpha}. \quad (8)$$

For $a_0 \neq 0$, we get from Eq. (7)

$$T(x_0) \leq \frac{1}{a_0(1-\alpha)} \ln \left(\frac{b_0}{b_0 - a_0 \epsilon^{1-\alpha}} \right). \quad (9)$$

Case 2. When $V(x_0) > \epsilon$: In this case we have

$$\begin{aligned} T(x_0) &\leq \int_0^{V(x_0)} \frac{dV}{-a_0 V + b_0 V^\alpha + c_0 V^{\beta-1}} \\ &\leq \int_0^\epsilon \frac{dV}{-a_0 V + b_0 V^\alpha + c_0 V^{\beta-1}} \\ &\quad + \int_\epsilon^\infty \frac{dV}{-a_0 V + b_0 V^\alpha + c_0 V^{\beta-1}} \end{aligned}$$

For, $a_0 > 0$, letting $z = \frac{V^{1-\alpha}}{\epsilon^{1-\alpha}}$ one obtains

$$\begin{aligned} &\int_\epsilon^\infty \frac{dV}{-a_0 V + b_0 V^\alpha + c_0 V^{\beta-1}} \\ &= \frac{1}{1-\alpha} \int_1^\infty \frac{dz}{-a_0 z + b_0 \epsilon^{\alpha-1} + c_0 \epsilon^\beta z^{\frac{\beta+1-\alpha}{1-\alpha}}} \\ &\leq \frac{1}{1-\alpha} \int_1^\infty \frac{dz}{(c_0 \epsilon^\beta - a_0) z^{\frac{\beta+1-\alpha}{1-\alpha}}} = \frac{1}{\beta(c_0 \epsilon^\beta - a_0)}. \end{aligned} \quad (10)$$

Further, for $a_0 \leq 0$ and letting $z = V^{1-\alpha}$ we get

$$\begin{aligned} &\int_\epsilon^\infty \frac{dV}{-a_0 V + b_0 V^\alpha + c_0 V^{\beta-1}} \\ &\leq \int_{\epsilon^{1-\alpha}}^\infty \frac{dz}{(1-\alpha) c_0 z^{\frac{\beta+1-\alpha}{1-\alpha}}} \\ &= \frac{1}{\beta c_0} \epsilon^{-\beta}. \end{aligned} \quad (11)$$

From (8), (9), (10) and (11), the settling time satisfies

$$\begin{aligned} T(x_0) &\leq T_{fix} \\ &= \begin{cases} \frac{1}{a_0(1-\alpha)} \ln \left(\frac{b_0}{b_0 - a_0 \epsilon^{1-\alpha}} \right) + \frac{1}{\beta c_0} \epsilon^{-\beta}, & a_0 < 0 \\ \frac{1}{b_0(1-\alpha)} \epsilon^{1-\alpha} + \frac{1}{\beta c_0} \epsilon^{-\beta}, & a_0 = 0 \\ \frac{1}{a_0(1-\alpha)} \ln \left(\frac{b_0}{b_0 - a_0 \epsilon^{1-\alpha}} \right) + \frac{1}{\beta(c_0 \epsilon^\beta - a_0)}, & a_0 > 0. \end{cases} \end{aligned}$$

This completes the proof of the Theorem 1. For some more clarification and insight into the proof, one can go through the Remark 3. $\square$

Remark 1 Lemma 7 of [17] is a particular case of the above theorem, which can be obtained by replacing $\epsilon = 1$. Further, it has to be noted that in most of the work reported in recent literature, more focus has been provided to converge the state trajectories as quickly as possible, but the upper bound of settling time used to be much larger than the actual convergence time which reduces the compatibility of the practical applications

and the theoretical results. Moreover, the selection of the parameters are generally done arbitrarily. Thus, we feel there should be certain relations/equality among the parameters; keeping these things in mind, let us dive into the next section.

3.2 On optimality condition

In this subsection, we will derive a condition for obtaining optimal value of the upper bound of the settling time obtained in Subsection 3.1 with respect to the parameter ϵ .

Case 1. For $a_0 = 0$, we have from Eq. (3)

$$T_{fix} = T_{fix}(\epsilon) = \frac{1}{b_0(1-\alpha)}\epsilon^{1-\alpha} + \frac{1}{\beta c_0}\epsilon^{-\beta}$$

and

$$\frac{dT_{fix}}{d\epsilon} = \frac{\epsilon^{-\alpha}}{b_0} - \frac{\epsilon^{-\beta-1}}{c_0}.$$

When $\frac{dT_{fix}}{d\epsilon} = 0$, one obtains $\epsilon = \left(\frac{b_0}{c_0}\right)^{\frac{1}{\beta+1-\alpha}}$.

Further

$$\left.\frac{d^2T_{fix}}{d\epsilon^2}\right|_{\epsilon=\left(\frac{b_0}{c_0}\right)^{\frac{1}{\beta+1-\alpha}}} = \epsilon^{-1} \left(\frac{-\alpha}{c_0\epsilon^{\beta+1}} + \frac{\beta+1}{c_0\epsilon^{\beta+1}} \right) > 0.$$

The minimum value of T_{fix} is therefore obtained when $\epsilon = \left(\frac{b_0}{c_0}\right)^{\frac{1}{\beta+1-\alpha}}$.

Case 2. For $a_0 < 0$, the value of T_{fix} is given by

$$T_{fix} = T_{fix}(\epsilon) = \frac{1}{a_0(1-\alpha)} \ln \left(\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}} \right) + \frac{1}{\beta c_0}\epsilon^{-\beta}$$

and

$$\frac{dT_{fix}}{d\epsilon} = \frac{\epsilon^{-\alpha}}{b_0 - a_0\epsilon^{1-\alpha}} - \frac{\epsilon^{-\beta-1}}{c_0},$$

which when equals to zero gives the implicit equation $\epsilon^{1-\alpha}(c_0\epsilon^\beta + a_0) = b_0$. Further,

$$\frac{d^2T_{fix}}{d\epsilon^2} = - \left(\frac{b_0\alpha\epsilon^{\alpha-1} - a_0}{(b_0\epsilon^\alpha - a_0\epsilon)^2} \right) + \frac{(\beta+1)\epsilon^{-\beta-2}}{c_0};$$

when ϵ satisfies $\epsilon^{1-\alpha}(c_0\epsilon^\beta + a_0) = b_0$, we get

$$\frac{d^2T_{fix}}{d\epsilon^2} = \frac{a_0(1-\alpha)}{(c_0\epsilon^{\beta+1})^2} + \frac{\beta+1-\alpha}{c_0\epsilon^{\beta+2}} > 0,$$

Therefore, the minimum value of T_{fix} is obtained at the solutions of the implicit equation $\epsilon^{1-\alpha}(c_0\epsilon^\beta + a_0) = b_0$.

Case 3. For $a_0 > 0$, T_{fix} is given by

$$T_{fix} = \frac{1}{a_0(1-\alpha)} \ln \left(\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}} \right) + \frac{1}{\beta(c_0\epsilon^\beta - a_0)},$$

and

$$\frac{dT_{fix}}{d\epsilon} = \frac{\epsilon^{-\alpha}}{b_0 - a_0\epsilon^{1-\alpha}} - \frac{c_0\epsilon^{\beta-1}}{(c_0\epsilon^\beta - a_0)^2}.$$

When $\frac{dT_{fix}}{d\epsilon} = 0$, we get $(c_0\epsilon^\beta - a_0)^2 = c_0\epsilon^{\beta-1}(b_0\epsilon^\alpha - a_0\epsilon)$. Now, when ϵ satisfies $(c_0\epsilon^\beta - a_0)^2 = c_0\epsilon^{\beta-1}(b_0\epsilon^\alpha - a_0\epsilon)$, one obtains

$$\frac{d^2T_{fix}}{d\epsilon^2} = b_0\epsilon^{\alpha-1}(1-\alpha) + (c_0\epsilon^\beta - a_0) \left\{ \beta + \frac{a_0\beta}{c_0\epsilon^\beta} \right\} > 0. \quad (12)$$

So, the minimum value of T_{fix} is obtained at the solution of the implicit equation $(c_0\epsilon^\beta - a_0)^2 = c_0\epsilon^{\beta-1}(b_0\epsilon^\alpha - a_0\epsilon)$.

Remark 2 Though it been shown that the solution of implicit equations corresponding to Case 2 and Case 3 provides the minimum values of T_{fix} , it is also essential to show that the solutions exist for both the equations. Further, the term “the minimum value of T_{fix} ” is coined in the sense of considering T_{fix} as a function of ϵ only. Now let us consider the following two parts to show the existence of solutions corresponding to the implicit equations presented in Case 2 and Case 3.

(i) We have the implicit equation $\epsilon^{1-\alpha}(c_0\epsilon^\beta + a_0) = b_0$ from Case 2.

Let $f_1(\epsilon) = \epsilon^{1-\alpha}(c_0\epsilon^\beta + a_0)$ and $f_2(\epsilon) = b_0$. It can be clearly seen that $f_1(\epsilon)$ decreases/increases as ϵ decreases/increases. And since $\epsilon \in (0, \infty)$ we can see that

$$\lim_{\epsilon \rightarrow 0} f_1(\epsilon) = 0 < b_0 = \lim_{\epsilon \rightarrow 0} f_2(\epsilon) \quad (13)$$

and

$$\lim_{\epsilon \rightarrow \infty} f_1(\epsilon) = \infty > b_0 = \lim_{\epsilon \rightarrow \infty} f_2(\epsilon) \quad (14)$$

Combining Eqs. (13) and (14) with the continuity of the function $f_3(\epsilon) = f_1(\epsilon) - f_2(\epsilon)$ we conclude

that there exists at least one solution of the equation $f_3(\epsilon) = 0$. Since $a_0 < 0$, the condition $a_0 < \min\{b_0, \frac{c_0}{2}, c_0\epsilon^\beta, b_0\epsilon^{-1}\}$ is satisfied for all $\epsilon > 0$.

(ii) From Case 3 one have $(c_0\epsilon^\beta - a_0)^2 = c_0\epsilon^{\beta-1}(b_0\epsilon^\alpha - a_0\epsilon)$.

Here since $a_0 > 0$, we have to find solution of the above equation restricted to the interval $\left(\left(\frac{a_0}{c_0}\right)^{\frac{1}{\beta}}, \left(\frac{b_0}{a_0}\right)\right)$.

Let $h(\epsilon) = (c_0\epsilon^\beta - a_0)^2 - c_0\epsilon^{\beta-1}(b_0\epsilon^\alpha - a_0\epsilon)$, which is a continuous function of ϵ and is defined for all $\epsilon > 0$. Now

$$\begin{aligned} h\left(\left(\frac{a_0}{c_0}\right)^{\frac{1}{\beta}}\right) &= a_0^2 - b_0c_0\left(\frac{a_0}{c_0}\right)^{1+\frac{\alpha-1}{\beta}} \\ &= a_0^2 - a_0b_0\left(\frac{a_0}{c_0}\right)^{\frac{\alpha-1}{\beta}} < a_0^2\left(1 - \left(\frac{a_0}{c_0}\right)^{\frac{\alpha-1}{\beta}}\right) \\ &< 0. \end{aligned}$$

and

$$\begin{aligned} h\left(\left(\frac{b_0}{a_0}\right)\right) &= c_0^2\left(\frac{b_0}{a_0}\right)^{2\beta} - a_0c_0\left(\frac{b_0}{a_0}\right)^\beta \\ &\quad - b_0c_0\left(\frac{b_0}{a_0}\right)^\beta\left(\frac{b_0}{a_0}\right)^{\alpha-1} + a_0^2 \\ &= c_0^2\left(\frac{b_0}{a_0}\right)^{2\beta} - a_0c_0\left(\frac{b_0}{a_0}\right)^\beta \\ &\quad - b_0c_0\left(\frac{b_0}{a_0}\right)^\beta\left(\frac{a_0}{b_0}\right)^{1-\alpha} + a_0^2 \\ &> c_0^2\left(\frac{b_0}{a_0}\right)^{2\beta} - 2b_0c_0\left(\frac{b_0}{a_0}\right)^\beta + a_0^2 \\ &= c_0\left(\frac{b_0}{a_0}\right)^\beta\left\{c_0\left(\frac{b_0}{a_0}\right)^\beta - 2b_0\right\} + a_0^2 \\ &> c_0\left(\frac{b_0}{a_0}\right)^\beta\left\{c_0\left(\frac{b_0}{a_0}\right)^\beta - 2b_0\right\} + a_0^2 \\ &> 0. \end{aligned} \tag{15}$$

Hence, the continuity of $h(\epsilon)$ along with the above two inequalities provides the sufficient condition for existence of at least one solution of $h(\epsilon)$ in the provided interval $\left(\left(\frac{a_0}{c_0}\right)^{\frac{1}{\beta}}, \left(\frac{b_0}{a_0}\right)\right)$.

Note: It should be noted that there may exist some other values of $\epsilon = \epsilon_*$ such that either $\epsilon_* \leq \left(\frac{a_0}{c_0}\right)^{\frac{1}{\beta}}$ or $\epsilon_* \geq \left(\frac{b_0}{a_0}\right)$ and $h(\epsilon_*) = 0$.

Now first suppose that, $\epsilon_* \leq \left(\frac{a_0}{c_0}\right)^{\frac{1}{\beta}}$ and $h(\epsilon_*) = 0$.

It simplifies to $a_0 \geq c_0\epsilon_*^\beta$, which is a contradiction to the assumption “ $a_0 < \min\{b_0, \frac{c_0}{2}, c_0\epsilon^\beta, b_0\epsilon^{-1}\}$ ” of Theorem 1. Similar situation arises for the other case, that is, for $\epsilon_* \geq \left(\frac{b_0}{a_0}\right)$ and $h(\epsilon_*) = 0$. So, only the existence of solution of $h(\epsilon) = 0$ is not enough but also it has to be within the open interval $\left(\left(\frac{a_0}{c_0}\right)^{\frac{1}{\beta}}, \left(\frac{b_0}{a_0}\right)\right)$ as it should satisfy the assumption “ $a_0 < \min\{b_0, \frac{c_0}{2}, c_0\epsilon^\beta, b_0\epsilon^{-1}\}$ ”. However, such type of restriction is not needed for the Case (ii) of above Remark 2 as $a_0 < 0$ and thus $a_0 < \min\{b_0, \frac{c_0}{2}, c_0\epsilon^\beta, b_0\epsilon^{-1}\}$ holds true for any $\epsilon > 0$.

Remark 3 In this remark, we are going to discuss the necessity of the condition “ $a_0 < \min\left\{b_0, \frac{c_0}{2}, c_0\epsilon^\beta, b_0\epsilon^{-1}\right\}$ ” mentioned in Theorem 1.

- $a_0 < b_0$ is used in Eq. (4), which is $-(b_0 - a_0)V^\alpha < 0$.
- $a_0 < \frac{c_0}{2}$ is used in Eq. (5), which is $-(c_0 - a_0)V^{\beta+1} < 0$. And also in obtaining Eq. (16) from Eq. (15), that is to show $c_0\left(\frac{b_0}{a_0}\right)^\beta\left\{c_0\left(\frac{b_0}{a_0}\right)^\beta - 2b_0\right\} + a_0^2$ greater than zero. It is worth mentioning here that, $a_0 < c_0$ would have been sufficient to obtain Eq. (5) in Theorem 1. But $a_0 < \frac{c_0}{2}$ is need in deducing Eq. (16) from Eq. (15). So, to fulfill both the requirements together we have considered $a_0 < \frac{c_0}{2}$.
- $a_0 < c_0\epsilon^\beta$ is used in 3rd part of Eq. (3). It is used to maintain the positivity of the term $\frac{1}{\beta(c_0\epsilon^\beta - a_0)}$ in $\frac{1}{a_0(1-\alpha)}\ln\left(\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}}\right) + \frac{1}{\beta(c_0\epsilon^\beta - a_0)}$. Thus one can obtain that violating the condition $a_0 < c_0\epsilon^\beta$ would have make the term $\frac{1}{\beta(c_0\epsilon^\beta - a_0)}$ undefined or may make the upper bound of the settling time $T_{fix} = \frac{1}{a_0(1-\alpha)}\ln\left(\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}}\right) + \frac{1}{\beta(c_0\epsilon^\beta - a_0)}$ negative, which is absurd. Apart from this, it has also been used in Eq. (12).
- $a_0 < b_0\epsilon^{-1}$ is used in Eq. (6), which is $a_0\epsilon - b_0 - c_0V^{\beta-1} < -c_0V^{\beta-1} < 0$. Apart from this it has also been used in 3rd part of Eq. (3). There we have

a term “ $\ln\left(\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}}\right)$ ” and thus it is not defined when $b_0 - a_0\epsilon^{1-\alpha} \leq 0$. To verify that this scenario never occurs let us dive into the following three cases depending on the values of ϵ :

Case 1: $\epsilon = 1$. Then, $b_0 - a_0\epsilon^{1-\alpha} = b_0 - a_0 > 0$ as, $a_0 < b_0$.

Case 2: $\epsilon > 1$. Then, $a_0\epsilon^{1-\alpha} < a_0\epsilon$ (as $0 < 1-\alpha < 1$). It thus implies that $b_0 - a_0\epsilon^{1-\alpha} > b_0 - a_0\epsilon > 0$ as ($a_0 < b_0\epsilon^{-1}$).

Case 3: $\epsilon < 1$. Then $b_0\epsilon^\alpha > b_0\epsilon$ (as $0 < \alpha < 1$). Also by virtue of $a_0 < b_0$ one can have $a_0\epsilon < b_0\epsilon$. Further we have $\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}} = \frac{b_0\epsilon^\alpha}{b_0\epsilon^\alpha - a_0\epsilon}$. Combining all these informations, one can have $b_0\epsilon^\alpha - a_0\epsilon > b_0\epsilon^\alpha - b_0\epsilon > 0$.

So, with the help of above three cases, we have $\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}} > 0$ and thus $\ln\left(\frac{b_0}{b_0 - a_0\epsilon^{1-\alpha}}\right)$ is always defined.

3.3 On accurate estimation of settling time

In this subsection of the manuscript, we will provide some results on accurate estimation of the settling time. More specifically, we will show that the difference between the true convergence time and the estimated upper bound of settling is smaller in case of optimal value of any ϵ than that of $\epsilon = 1$. And thus will show that the estimated upper bound of the settling time of this manuscript is more accurate than that of [17].

Corollary 1 *Under the following Lyapunov equality, the upper bound estimate of the settling time is more accurate for the case of the optimal value of ϵ than that of $\epsilon = 1$, which is considered in the article [17].*

Proof Considering the equality of Eq. (2) of Theorem 1 we have,

$$\frac{d}{dt}V(x) = a_0V(x) - b_0(V(x))^\alpha - c_0(V(x))^{\beta+\text{sign}(V(x)-\epsilon)}.$$

Which can be written as

$$\begin{cases} \frac{d}{dt}V(x) = a_0V(x) - b_0(V(x))^\alpha - c_0(V(x))^{\beta-1}, & V(x) < \epsilon \\ \frac{d}{dt}V(x) = a_0V(x) - b_0(V(x))^\alpha - c_0(V(x))^{\beta+1}, & V(x) > \epsilon \end{cases} \quad (17)$$

Table 1 Case 1 of Corollary 1

	$\epsilon = \epsilon_0$	$\epsilon = 1$
$\dot{V} =$	$-(V(x))^{\beta-1}$	$-(V(x))^{\beta+1}$

Now, it is the third term $((V(x))^{\beta+\text{sign}(V(x)-\epsilon)})$ which makes all the differences. Hence, we will only concentrate on this. Thus, let us consider:

$$\begin{cases} \frac{d}{dt}V(x) = \dot{V} = -(V(x))^{\beta-1}, & V(x) < \epsilon \\ \frac{d}{dt}V(x) = \dot{V} = -(V(x))^{\beta+1}, & V(x) > \epsilon \end{cases} \quad (18)$$

Now, let $\epsilon = \epsilon_0 \neq 1$ be any given constant. We will study above equation in two cases, corresponding to $\epsilon = 1$ and $\epsilon = \epsilon_0$. **Note that here we have not considered ϵ_0 to be optimal.**

Case 1: $\epsilon_0 > 1$. Let $V(x_0) = V_0$ be the initial condition. We break the interval $(0, \infty)$ into three parts as

- (i) $V \in (0, 1]$.
- (ii) $V \in (1, \epsilon_0)$.
- (iii) $V \in [\epsilon_0, \infty)$.

Note that, $\dot{V}$ acquires same convergence rate for $\epsilon = \epsilon_0$ and $\epsilon = 1$ corresponding to both the conditions (i) and (iii). Hence, there is same rate of convergence of the trajectories when V is in these two intervals. But, when $V \in (1, \epsilon_0)$ we have following table (Table 1).

From the above Table 1, it is clear that the convergence speed corresponding to $\epsilon = 1$ is greater than that of $\epsilon = \epsilon_0$. So, the trajectory will converge faster in case of $\epsilon = 1$ than that of $\epsilon = \epsilon_0$, i.e., the true convergence time in case of $\epsilon = 1$ is lesser than that of $\epsilon = \epsilon_0$. Moreover, if we consider $V_0 \in (0, 1]$, then the case $x \in (1, \epsilon_0)$ does not come into account and we will have same rate of convergence for both $\epsilon = 1$ and $\epsilon = \epsilon_0$, i.e., the true convergence time in both the cases ($\epsilon = 1$ and $\epsilon = \epsilon_0$) will be same.

Case 2: $\epsilon_0 < 1$. Let $V(x_0) = V_0$ be the initial condition. Similar to **Case 1**, we break the interval $(0, \infty)$ into three parts as

- (i) $V \in (0, \epsilon_0]$.
- (ii) $V \in (\epsilon_0, 1)$.
- (iii) $V \in [1, \infty)$.

Similar to above discussion we have, $\dot{V}$ acquires same convergence rate for $\epsilon = \epsilon_0$ and $\epsilon = 1$ corresponding

Table 2 Case 2 of Corollary 1

	$\epsilon = \epsilon_0$	$\epsilon = 1$
$\dot{V} =$	$-(V(x))^{\beta+1}$	$-(V(x))^{\beta-1}$

to both the conditions (i) and (iii). Hence, there is same rate of convergence of the trajectories when V is in these two intervals. But, when $V \in (\epsilon_0, 1)$ we have following table (Table 2).

Note that $V < 1$, and so $(V(x))^{\beta+1} < (V(x))^{\beta-1}$. Thus, from the above Table 2 one can obtain that the convergence rate in case $\epsilon = 1$ is greater than that of $\epsilon = \epsilon_0$, i.e., the true convergence time in case of $\epsilon = 1$ is lesser than that of $\epsilon = \epsilon_0$. Meanwhile, if we consider $V_0 \in (0, \epsilon_0]$, then the case $V \in (\epsilon_0, 1)$ does not come into account and we will have same rate of convergence for both $\epsilon = 1$ and $\epsilon = \epsilon_0$ i.e., the true convergence time in both the cases ($\epsilon = 1$ and $\epsilon = \epsilon_0$) will be same. Thus from **Case 1 and case 2**, one can conclude that the convergence rate of $\epsilon = 1$ is greater or equal to that of $\epsilon = \epsilon_0$. Specifically, if we consider ϵ_0 as the optimal value, then the upper bound of the settling time is less than that for $\epsilon = 1$ (see Subsection 3.2). Combining above two sentences one can conclude that the upper bound estimate of the settling time is more accurate for the case of the optimal value of any ϵ than that of $\epsilon = 1$. Please refer to Fig. 1 for more clarification. This completes the proof. $\square$

Remark 4 It should be noted that the optimal value of ϵ may not reduce the actual convergence time in comparison to $\epsilon = 1$; instead, it helps to reduce the difference between the true convergence time and the upper bound estimate of the settling time, i.e., the upper bound estimate of the settling time is more accurate in case of the optimal value of ϵ than that of $\epsilon = 1$. Thus, it maximizes the validity of the theoretical results in practical applications. Further, the numerical justification of this subsection is shown in Remark 5 of Sect. 6. Also note that we have not mentioned the case $V(x) = \epsilon$ explicitly in Eqs. (17) and (18) as it will not affect our conclusions of the Corollary.

4 System control

In this section, we will be designing the sliding surface and the controllers corresponding to the error system

obtained from difference of state trajectories of a master and a slave system.

Let us consider the master system as

$$\begin{cases} \dot{x}_{01} = f_1(x) \\ \dot{x}_{02} = f_2(x) \\ \vdots \\ \dot{x}_{0n} = f_n(x) \end{cases} \quad (19)$$

where $x = [x_{01}, x_{02}, \dots, x_{0n}]^T \in \mathbb{R}^n$ is the system states.

Corresponding slave system can be modeled as

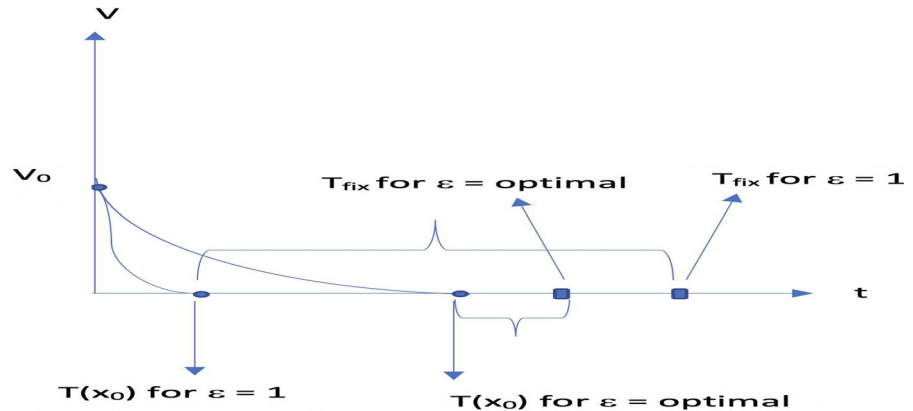
$$\begin{cases} \dot{y}_{01} = g_1(y) + \Delta g_1(t, y) + u_1 \\ \dot{y}_{02} = g_2(y) + \Delta g_2(t, y) + u_2 \\ \vdots \\ \dot{y}_{0n} = g_n(y) + \Delta g_n(t, y) + u_n \end{cases} \quad (20)$$

where $y = [y_{01}, y_{02}, \dots, y_{0n}]^T \in \mathbb{R}^n$ is the observer state, $\Delta g_1(t, y), \Delta g_2(t, y), \dots, \Delta g_n(t, y)$ are the disturbances bounded by positive constants $\delta_1, \delta_2, \delta_3$ respectively i.e., $|\Delta g_1(t, y)| < \delta_1, |\Delta g_2(t, y)| < \delta_2, \dots, |\Delta g_n(t, y)| < \delta_n$ and $u_1, u_2, \dots, u_n$ are the controllers. The error states are defined as $e_{01} = y_{01} - x_{01}; e_{02} = y_{02} - x_{02}; \dots; e_{0n} = y_{0n} - x_{0n}$. Then the error system is described as follows:

$$\begin{cases} \dot{e}_{01} = g_1(y) + \Delta g_1(t, y) - f_1(x) + u_1 \\ \dot{e}_{02} = g_2(y) + \Delta g_2(t, y) - f_2(x) + u_2 \\ \vdots \\ \dot{e}_{0n} = g_n(y) + \Delta g_n(t, y) - f_n(x) + u_n \end{cases} \quad (21)$$

4.1 Sliding surface designing

To stabilize the error system (21) within a fixed time of the form of Theorem 1, let us design NSTSM as

Fig. 1 For Corollary 1

$$S_{0i} = e_{0i} + \int_0^t \left\{ -\frac{a_0}{2} e_{0i} + \frac{b_0}{2^{\frac{1+\alpha}{2}}} |e_{0i}|^\alpha \operatorname{sign}(e_{0i}) + \frac{c_0 n^{\beta_s}}{2^{\beta_s+1}} |e_{0i}|^{2\beta_s+1} \operatorname{sign}(e_{0i}) \right\} d\tau \quad (22)$$

where $i = 1, 2, \dots, n$, $\beta_s = \beta + \operatorname{sign}(V - \epsilon)$ and $a_0, b_0, c_0, \alpha, \beta, \epsilon$ are as provided in Theorem 1.

Theorem 2 Once the state trajectories of the system (21) reaches the NSTSM (22), it will converge to zero within some fixed-time. And the settling time is given by:

$$T_1 \leq \begin{cases} \frac{1}{a_0(1-\alpha_k)} \ln \left(\frac{b_0}{b_0 - a_0 \epsilon^{1-\alpha_k}} \right) + \frac{\epsilon^{-\beta_k}}{c_0 \beta_k}, & a_0 < 0 \\ \frac{1}{b_0(1-\alpha_k)} \epsilon^{1-\alpha_k} + \frac{\epsilon^{-\beta_k}}{c_0 \beta_k}, & a_0 = 0 \\ \frac{1}{a_0(1-\alpha_k)} \ln \left(\frac{b_0}{b_0 - a_0 \epsilon^{1-\alpha_k}} \right) + \frac{1}{\beta_k(c_0 \epsilon^{\beta_k - a_0})}, & a_0 > 0 \end{cases},$$

where $\alpha_k = \frac{\alpha+1}{2}$ and $\beta_k = \beta + 1$.

Proof When the error system (21) is on the sliding surface $S(t) = \{e_{0i} \mid S_{0i}(e) = 0\}$ then $\dot{S}_{0i} = 0$. Therefore

$$\dot{e}_{0i} = \frac{a_0}{2} e_{0i} - \frac{b_0}{2^{\frac{1+\alpha}{2}}} |e_{0i}|^\alpha \operatorname{sign}(e_{0i}) - \frac{n^{\beta_s}}{2^{\beta_s+1}} |e_{0i}|^{2\beta_s+1} \operatorname{sign}(e_{0i}), \quad i = 1, 2, \dots, n$$

Considering the Lyapunov function $V = \frac{1}{2}(e_{o1}^2 + e_{o2}^2 + \dots + e_{on}^2)$, we obtain

$$\dot{V} = e_{o1}\dot{e}_{o1} + e_{o2}\dot{e}_{o2} + \dots + e_{on}\dot{e}_{on}$$

$$\begin{aligned} &= \frac{a_0}{2} (e_{o1}^2 + e_{o2}^2 + \dots + e_{on}^2) \\ &\quad - \frac{b_0}{2^{\frac{1+\alpha}{2}}} (|e_{o1}|^{\alpha+1} + |e_{o2}|^{\alpha+1} + \dots + |e_{on}|^{\alpha+1}) \\ &\quad - \frac{c_0 n^{\beta_s}}{2^{\beta_s+1}} (|e_{o1}|^{2\beta_s+2} + |e_{o2}|^{2\beta_s+2} + \dots + |e_{on}|^{2\beta_s+2}) \\ &= a_0 V - b_0 \left\{ \left(\frac{e_{o1}^2}{2} \right)^{\frac{\alpha+1}{2}} + \left(\frac{e_{o2}^2}{2} \right)^{\frac{\alpha+1}{2}} + \dots + \left(\frac{e_{on}^2}{2} \right)^{\frac{\alpha+1}{2}} \right\} \\ &\quad - c_0 n^{\beta_s} \left\{ \left(\frac{e_{o1}^2}{2} \right)^{\beta_s+1} + \left(\frac{e_{o2}^2}{2} \right)^{\beta_s+1} + \dots + \left(\frac{e_{on}^2}{2} \right)^{\beta_s+1} \right\} \end{aligned}$$

Then, Lemma 1 gives

$$\begin{aligned} \dot{V} &\leq a_0 V - b_0 \left(\frac{e_{o1}^2 + e_{o2}^2 + \dots + e_{on}^2}{2} \right)^{\frac{\alpha+1}{2}} \\ &\quad - c_0 \left(\frac{e_{o1}^2 + e_{o2}^2 + \dots + e_{on}^2}{2} \right)^{\beta_s+1} \\ &= a_0 V - b_0 V^{\frac{\alpha+1}{2}} - c_0 V^{\beta_s+1} \\ &= a_0 V - b_0 V^{\frac{\alpha+1}{2}} - c_0 V^{(\beta+1)+\operatorname{sign}(V-\epsilon)} \\ &= a_0 V - b_0 V^{\alpha_k} - c_0 V^{\beta_k+\operatorname{sign}(V-\epsilon)}. \end{aligned}$$

By Theorem 1, we can conclude that the state trajectories of system (21) will converge to zero within the fixed-time T_1 once it reaches the sliding surface (22). $\square$

4.2 Controller design

To synchronize the systems (19) and (20) in a fixed-time, we choose Eq. (22) as the sliding surface. Then, let us construct the control input as

$$u_i = -h_i(e) - \delta_i \operatorname{sign}(S_{0i}) - \left\{ -\frac{a_1}{2} S_{0i} + \frac{b_1}{2^{\frac{1+\alpha}{2}}} |S_{0i}|^{\alpha_1} \operatorname{sign}(S_{0i}) + \frac{c_1 n^{\beta_0}}{2^{\beta_0+1}} |S_{0i}|^{2\beta_0+1} \operatorname{sign}(S_{0i}) \right\}, i = 1, 2, \dots, n \quad (23)$$

where

$$h_i(e) = g_i(y) - f_i(x) - \frac{a_0}{2} e_{0i} + \frac{b_0}{2^{\frac{1+\alpha}{2}}} |e_{0i}|^{\alpha} \operatorname{sign}(e_{0i}) + \frac{c_0 n^{\beta_s}}{2^{\beta_s+1}} |e_{0i}|^{2\beta_s+1} \operatorname{sign}(e_{0i}) \quad (24)$$

$\alpha_1, \beta_1, \epsilon_1, a_1, b_1, c_1$ are real constants satisfying $0 < \alpha_1 < 1, \beta_1 > 1, b_1, c_1 > 0$ and $a_1 < \min \left\{ b_1, \frac{c_1}{2}, c_1 \epsilon_1^{\beta_1}, b_1 \epsilon_1^{-1} \right\}$. We define V_S as the Lyapunov function and assume that β_0 satisfies $\beta_0 = \beta_1 + \operatorname{sign}(V_S - \epsilon_1)$.

Theorem 3 Using the controller (23), the error system (21) will reach the sliding surface (22) in fixed-time, given by

$$T_2 \leq \begin{cases} \frac{1}{a_1(1-\alpha_u)} \ln \left(\frac{b_1}{b_1 - a_1 \epsilon_1^{1-\alpha_u}} \right) + \frac{\epsilon_1^{-\beta_u}}{c_1 \beta_u}, & a_0 < 0 \\ \frac{1}{b_1(1-\alpha_u)} \epsilon_1^{1-\alpha_u} + \frac{\epsilon_1^{-\beta_u}}{c_1 \beta_u}, & a_0 = 0 \\ \frac{1}{a_1(1-\alpha_u)} \ln \left(\frac{b_1}{b_1 - a_1 \epsilon_1^{1-\alpha_u}} \right) + \frac{1}{\beta_u(c_1 \epsilon_1^{\beta_u} - a_1)}, & a_0 > 0 \end{cases},$$

where $\alpha_u = \frac{\alpha_1+1}{2}$ and $\beta_u = \beta_1 + 1$.

Proof Differentiating the Lyapunov function $V_S = \frac{1}{2} (S_{01}^2 + S_{02}^2 + \dots + S_{0n}^2)$ along the trajectories of error system (21), we get

$$\begin{aligned} \dot{V}_S &= S_{01} \dot{S}_{01} + S_{02} \dot{S}_{02} + \dots + S_{0n} \dot{S}_{0n} \\ &= \sum_{l=1}^n S_{0l} \left(\dot{e}_l - \frac{a_0}{2} e_{0l} + \frac{b_0}{2^{\frac{1+\alpha}{2}}} |e_{0l}|^{\alpha} \operatorname{sign}(e_{0l}) + \frac{c_0 n^{\beta_s}}{2^{\beta_s+1}} |e_{0l}|^{2\beta_s+1} \operatorname{sign}(e_{0l}) \right) \end{aligned}$$

Replacing the values of $\dot{e}_i$ from Eq. (21) and using the controllers u_i , one can obtain

$$\begin{aligned} \dot{V}_S &= \frac{a_1}{2} (S_{01}^2 + S_{02}^2 + \dots + S_{0n}^2) - \frac{b_1}{2^{\frac{\alpha_1+1}{2}}} (|S_{01}|^{\alpha_1+1} + |S_{02}|^{\alpha_1+1} + \dots + |S_{0n}|^{\alpha_1+1}) \\ &\quad - \frac{c_1 n^{\beta_0}}{2^{\beta_0+1}} (|S_{01}|^{2\beta_0+1} + |S_{02}|^{2\beta_0+1} + |S_{03}|^{2\beta_0+1}) - \sum_{l=1}^3 (\delta_l \operatorname{sign}(S_{0l}) + \Delta g_l(y)) S_{0l} \\ &\leq a_1 V_S - b_1 \left\{ \left(\frac{S_{01}^2}{2} \right)^{\frac{\alpha_1+1}{2}} + \left(\frac{S_{02}^2}{2} \right)^{\frac{\alpha_1+1}{2}} + \dots + \left(\frac{S_{0n}^2}{2} \right)^{\frac{\alpha_1+1}{2}} \right\} - c_1 n^{\beta_0} \left\{ \left(\frac{S_{01}^2}{2} \right)^{\beta_0+1} + \left(\frac{S_{02}^2}{2} \right)^{\beta_0+1} + \dots + \left(\frac{S_{0n}^2}{2} \right)^{\beta_0+1} \right\}. \end{aligned}$$

Now, in view of Lemma 1, we get

$$\begin{aligned}\dot{V}_S &\leq a_1 V_S - b_1 \left(\frac{S_{01}^2 + S_{02}^2 + \dots S_{0n}^2}{2} \right)^{\frac{\alpha_1+1}{2}} \\ &\quad - c_1 \left(\frac{S_{01}^2 + S_{02}^2 + \dots + S_{0n}^2}{2} \right)^{\beta_0+1} \\ &= a_1 V_S - b_1 V_S^{\frac{\alpha_1+1}{2}} - c_1 V_S^{\beta_0+1} = a_1 V_S \\ &\quad - b_1 V_S^{\frac{\alpha_1+1}{2}} - c_1 V_S^{(\beta_1+1)+\text{sign}(V_S-\epsilon_1)} \\ &= a_1 V_S - b_1 V_S^{\alpha_u} - c_1 V_S^{\beta_u+\text{sign}(V_S-\epsilon_1)}\end{aligned}$$

Thus, in accordance to Theorem 1, the error system will reach the sliding surface $S(t) = 0$ within some fixed-time T_2 . This completes the proof of the Theorem 3. Combining Theorem 2 and Theorem 3, we get that the error trajectories will converge to zero within some fixed time $T_0 = T_1 + T_2$. $\square$

5 Application in secure communication

This section of the article is dedicated to the application of the previously obtained results in the field of secure communication. A communication scheme depicting the transmission of signals between master and slave system is shown in Fig. 2.

The 3D-chaotic system proposed in the paper [41] is considered here, which has been verified to be more complex as compared to many chaotic systems present in the literature. Hence, the master system can be modeled as

$$\begin{cases} \dot{x}_1 = (\delta - 1)x_2 - \delta x_1 - x_3 \\ \dot{x}_2 = (r + 1)x_1 - (1 - a)x_2 - px_1x_3 \\ \dot{x}_3 = qx_1x_2 - bx_3 + x_1x_3 - cx_1 \end{cases} \quad (25)$$

where $x = [x_1, x_2, x_3]^T \in \mathbb{R}^3$ is the system state, and δ, r, a, p, q, b, c constant parameters. The mentioned chaotic system has demonstrated a chaotic behavior for several choices of δ, r, a, p, q, b, c . Here we will be considering the values of these parameters as $\delta = 33, r = 45.6, a = 12, p = q = 10, b = 5, c = 11$. Fig 3 shows the chaotic behavior of the system (25).

Now let corresponding slave system be modeled as

$$\begin{cases} \dot{y}_1 = (\delta - 1)y_2 - \delta y_1 - y_3 + u_1 \\ \dot{y}_2 = (r + 1)z - (1 - a)y_2 - pz y_3 + u_2 \\ \dot{y}_3 = qz y_2 - by_3 + zy_3 - cz + u_3 \end{cases} \quad (26)$$

where $z = x_{01} + m$ is the transmitted signal which is transmitted from the master system to the slave system, m is the message signal which has been masked by the x_{01} and $y = [y_1, y_2, y_3]^T \in \mathbb{R}^3$ is the observer state. Further, let the signal be transmitted through a noise channel, so that the noises will be introduced in the error system. Hence let $\Delta e_{01}, \Delta e_{02}, \Delta e_{03}$ are the noises bounded by positive constants $\delta_1, \delta_2, \delta_3$ respectively i.e., $|\Delta e_{01}| < \delta_1, |\Delta e_{02}| < \delta_2, |\Delta e_{03}| < \delta_3$ and u_1, u_2, u_3 are the controllers. The aforementioned communication scheme is depicted through Fig. 2.

The error system between the master and slave system is defined as $e_{01} = y_1 - x_{01}; e_{02} = y_2 - x_{02}; e_{03} = y_3 - x_{03}$.

Then the error system is described as follows:

$$\begin{cases} \dot{e}_{01} = (\delta - 1)e_{02} - \delta e_{01} - e_{03} + \Delta e_{01} + u_1 \\ \dot{e}_{02} = (r + 1)(z - x_{01}) - (1 - a)e_{02} - pz y_3 \\ \quad + px_{01}x_{03} + \Delta e_{02} + u_2 \\ \dot{e}_{03} = q(z y_2 - x_{01}x_{02}) - be_{03} + zy_{03} \\ \quad - x_{01}x_{03} - c(z - x_{01}) + \Delta e_{03} + u_3 \end{cases} \quad (27)$$

Now comparing Eqs. (21) and (27), one can obtain

$$\begin{aligned}g_1(y) - f_1(x) &= (\delta - 1)e_{02} - \delta e_{01} - e_{03}; g_2(y) - f_2(x) \\ &= (r + 1)(z - x_{01}) - (1 - a)e_{02} \\ &\quad - pz y_3 + px_{01}x_{03}; \\ g_3(y) - f_3(x) &= q(z y_2 - x_{01}x_{02}) \\ &\quad - be_{03} + zy_{03} - x_{01}x_{03} - c(z - x_{01}),\end{aligned}$$

and

$$\begin{aligned}\Delta g_1(t, y) &= \Delta e_{01}; \Delta g_2(t, y) = \Delta e_{02}; \Delta g_3(t, y) \\ &= \Delta e_{03}.\end{aligned}$$

Then, constructing the sliding surface and the controllers, for $n = 3$, as described in Sect. 4, we can stabilize the error system. Further, the specific form of the controllers are given as follows:

$$u_1 = - \left\{ (\delta - 1)e_{02} - \delta e_{01} - e_{03} - \frac{a_0}{2}e_{01} \right.$$

$$\begin{aligned}
& + \frac{b_0}{2^{\frac{1+\alpha}{2}}} |e_{01}|^\alpha \operatorname{sign}(e_{01}) \\
& + \frac{c_0 n^{\beta_s}}{2^{\beta_s+1}} |e_{01}|^{2\beta_s+1} \operatorname{sign}(e_{01}) \Big\} \\
& - \delta_1 \operatorname{sign}(S_{01}) - \left\{ -\frac{a_1}{2} S_{01} + \frac{b_1}{2^{\frac{1+\alpha_1}{2}}} |S_{01}|^{\alpha_1} \right. \\
& \left. \operatorname{sign}(S_{01}) + \frac{c_1 n^{\beta_0}}{2^{\beta_0+1}} |S_{01}|^{2\beta_0+1} \operatorname{sign}(S_{01}) \right\}; \\
u_2 = & - \left\{ (r+1)(z-x_{01}) - (1-a)e_{02} - pz y_{03} \right. \\
& + px_{01} x_{03} - \frac{a_0}{2} e_{02} + \frac{b_0}{2^{\frac{1+\alpha}{2}}} |e_{02}|^\alpha \operatorname{sign}(e_{02}) \\
& + \frac{c_0 n^{\beta_s}}{2^{\beta_s+1}} |e_{02}|^{2\beta_s+1} \operatorname{sign}(e_{02}) \Big\} \\
& - \delta_2 \operatorname{sign}(S_{02}) - \left\{ -\frac{a_1}{2} S_{02} + \frac{b_1}{2^{\frac{1+\alpha_1}{2}}} |S_{02}|^{\alpha_1} \right. \\
& \left. \operatorname{sign}(S_{02}) + \frac{c_1 n^{\beta_0}}{2^{\beta_0+1}} |S_{02}|^{2\beta_0+1} \operatorname{sign}(S_{02}) \right\}; \\
u_3 = & - \{ q(z y_{02} - x_{01} x_{02}) - b e_{03} + z y_{03} - x_{01} x_{03} \\
& - c(z-x_{01}) - \frac{a_0}{2} e_{03} + \frac{b_0}{2^{\frac{1+\alpha}{2}}} |e_{03}|^\alpha \operatorname{sign}(e_{03}) \\
& + \frac{c_0 n^{\beta_s}}{2^{\beta_s+1}} |e_{03}|^{2\beta_s+1} \operatorname{sign}(e_{03}) \Big\} \\
& - \delta_3 \operatorname{sign}(S_{03}) - \left\{ -\frac{a_1}{2} S_{03} + \frac{b_1}{2^{\frac{1+\alpha_1}{2}}} |S_{03}|^{\alpha_1} \right. \\
& \left. \operatorname{sign}(S_{03}) + \frac{c_1 n^{\beta_0}}{2^{\beta_0+1}} |S_{03}|^{2\beta_0+1} \operatorname{sign}(S_{03}) \right\};
\end{aligned}$$

where S_{01} , S_{02} , S_{03} are the sliding surfaces given in Eq. (22) and the parameters satisfy the conditions provided in Sect. 4. Hence, at the receiver end, the states of the master and slave systems are completely synchronized in fixed time i.e., $x_{0i} = y_{0i}$, and the message signal m can be retrieved as $z - x_{01} = m \approx \tilde{m} = z - y_{01}$. The claims are verified in Sect. 6.

6 Numerical verification and discussion.

This section provides four examples to verify the truthfulness of the results proposed in previous sections (Fig. 3).

Simulation 1: In this simulation, we will verify the efficacy of the assumptions and conditions used in Theorem 1. For this, let us consider the following example:

$$\begin{aligned}
\dot{x}(t) = & \frac{a_0}{2} x(t) - \frac{b_0}{2} |x(t)|^{\frac{1}{3}} \operatorname{sign}(x(t)) \\
& - \frac{c_0}{2} |x(t)|^{3+2\operatorname{sign}(x^2(t)-1)}. \quad (28)
\end{aligned}$$

Let us further consider the Lyapunov function $V(x) = x^2$. Then, in combination with the Eq. (28), one can get

$$\begin{aligned}
\dot{V} = & a_0 x^2(t) - b_0 |x(t)|^{\frac{4}{3}} - c_0 |x(t)|^{4+2\operatorname{sign}(x^2(t)-1)} \\
= & a_0 V - b_0 V^{\frac{2}{3}} - c_0 V^{2+\operatorname{sign}(V-1)}. \quad (29)
\end{aligned}$$

Comparing Eq. (29) with Eq. (2) one can obtain, $\alpha = \frac{2}{3}$, $\beta = 2$ and $\epsilon = 1$. Now, we will consider two sets of parameters: (i) satisfying the condition $a_0 < \min \{b_0, \frac{c_0}{2}, c_0 \epsilon^\beta, b_0 \epsilon^{-1}\}$, that is, $a_0 < \min \{b_0, \frac{c_0}{2}\}$ and (ii) violating the condition $a_0 < \min \{b_0, \frac{c_0}{2}, c_0 \epsilon^\beta, b_0 \epsilon^{-1}\}$, that is, $a_0 \geq \min \{b_0, \frac{c_0}{2}\}$. For example consider,

(i) $b_0 = 4, c_0 = 5, a_0 = 2$. That is, $a_0 < \min \{b_0, \frac{c_0}{2}\}$.

(ii) $b_0 = 4, c_0 = 4, a_0 = 10$. That is, $a_0 \geq \min \{b_0, \frac{c_0}{2}\}$.

From Fig. 4 one can obtain that, parameters violating the condition $a_0 < \min \{b_0, \frac{c_0}{2}, c_0 \epsilon^\beta, b_0 \epsilon^{-1}\}$ may not converge to the origin. Whereas, parameters satisfying the condition $a_0 < \min \{b_0, \frac{c_0}{2}, c_0 \epsilon^\beta, b_0 \epsilon^{-1}\}$ converges to the origin within the fixed-time. This validates the efficacy of the condition.

Simulation 2: In this simulation, we will demonstrate the performance of the constructed sliding surface corresponding to several sets of parameters. We have the sliding surface as

$$\begin{aligned}
S_0 = & x + \int_0^t \left\{ -ax + b |x|^\alpha \operatorname{sign}(x) \right. \\
& \left. + c |x|^{\beta+\operatorname{sign}(|x|-\epsilon)} \operatorname{sign}(x) \right\} d\tau = 0, \quad (30)
\end{aligned}$$

where real constants $\alpha, \beta, \epsilon, a, b, c$ satisfies $0 < \alpha < 1$, $\beta > 1$, $b, c > 0$ and $a < \min \{b, \frac{c}{2}, c \epsilon^\beta, b \epsilon^{-1}\}$. To demonstrate the performance of the sliding surface, we will consider $a < 0$ and the corresponding four different cases of the sliding surface parameters with the initial state $x(0) = 3$: Case 1. $a = -1, b = c = 1, \alpha = \frac{5}{9}, \beta = \frac{9}{5}, \epsilon = 1$; Case 2. $a = -2, b = c = 2, \alpha = \frac{5}{9}, \beta = \frac{9}{5}, \epsilon = 1$; Case 3. $a = -1, b = c =$

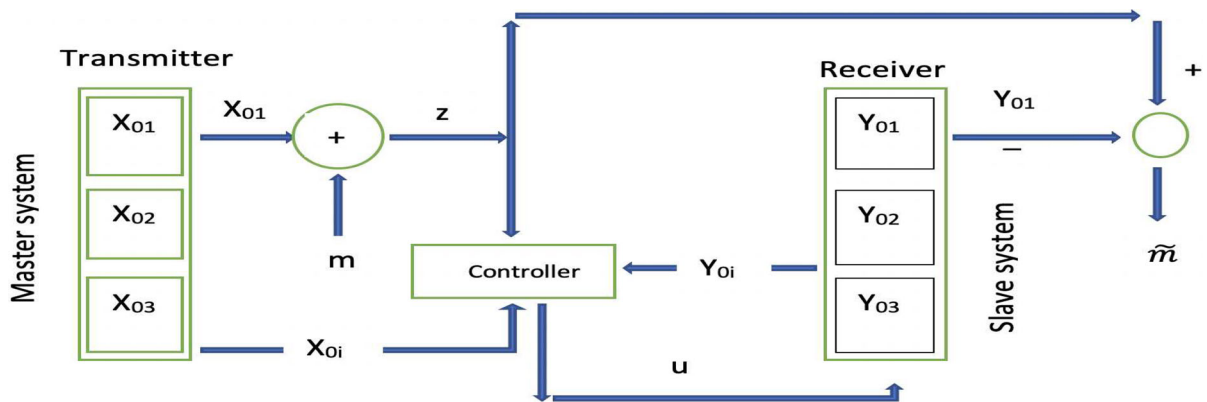


Fig. 2 Communication Scheme

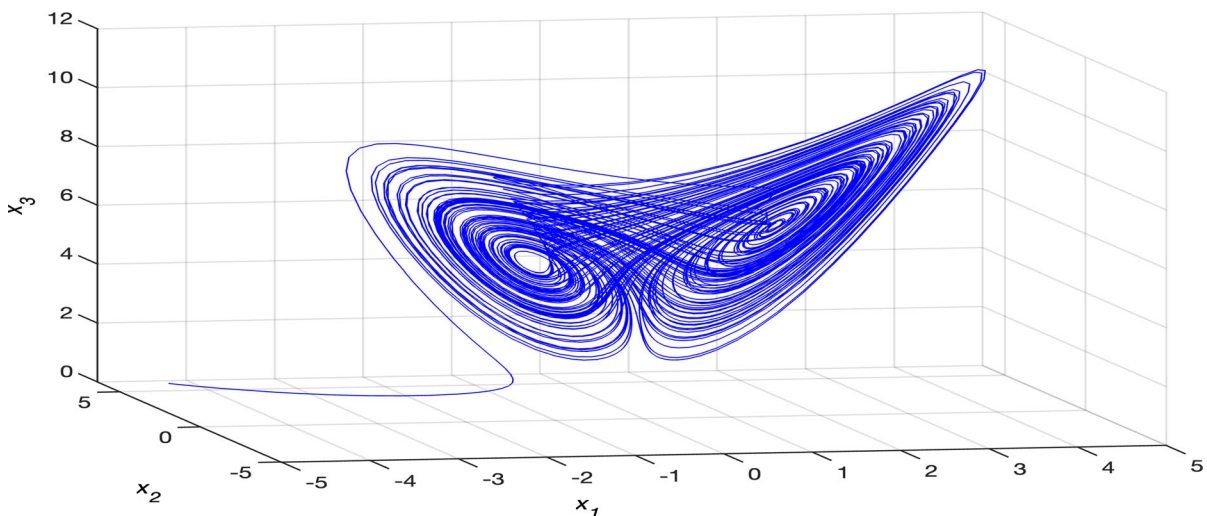


Fig. 3 Chaotic trajectories of (25) with initial condition $(-5, 2, 1)$

1, $\alpha = \frac{5}{9}$, $\beta = \frac{11}{5}$, $\epsilon = 1$; and Case 4. $a = -1$, $b = c = 1$, $\alpha = \frac{7}{9}$, $\beta = \frac{9}{5}$, $\epsilon = 1$. Fig. 5 shows the variation in convergence rate of the corresponding sliding surface S_0 with respect to variation in its parameters. From Case 1 and Case 2, it can be obtained that greater the values of $-a, b, c$ faster be the convergence rate. Comparing Case 1 with Case 3, one can obtain that greater the value of β faster be the convergence rate when the state x is away from origin. By the comparison of Case 1 and Case 4, one can conclude that smaller the value of α , faster is the convergence rate when the state x is nearer to the origin. The variation in convergence rate of the state x of the sliding surface S_0 with respect to the parameter ϵ is provided separately in Remark 5.

Remark 5 Regardless of the optimal value of ϵ , $\epsilon = 1$ will give the faster or equal rate of convergence than

that of any other values of ϵ corresponding to same set of other parameters of the Eq. (30) depending on the same absolute value of the initial condition to be in $(\min\{1, \epsilon\}, \infty)$ or $(0, \min\{1, \epsilon\}]$ respectively. But, as noted earlier we are also focusing on more accurate estimation of the settling time rather than only to minimize the convergence time. To demonstrate this let us consider three cases of values of a .

Case 1. For $a = 0$. Let us consider the parameters as $b = 4, c = 1, \beta = \frac{3}{2}, \alpha = \frac{1}{2}$. Then by Case 1 of optimality condition we have $\epsilon = 2$. And thus the upper bound of the settling time as per Theorem 1 is given by $T_{fix} \approx 0.943$, whereas for $\epsilon = 1$ the upper bound is given by $T_{max} \approx 1.166$. Hence we get $T_{fix} < T_{max}$. Fig. 6 clearly demonstrates this fact.

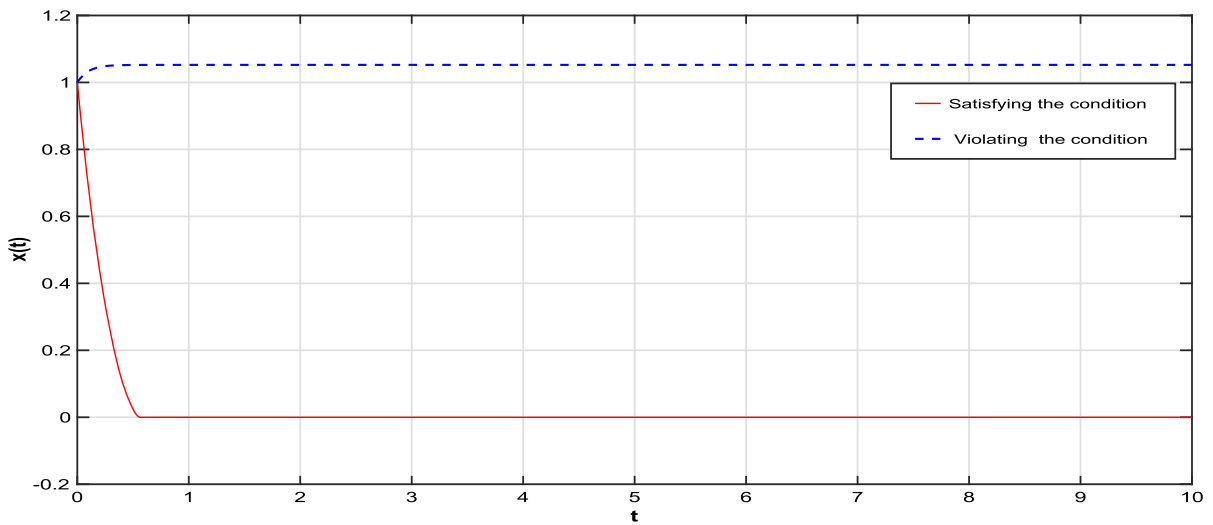


Fig. 4 State trajectory of (28) with conditions (i) and (ii)

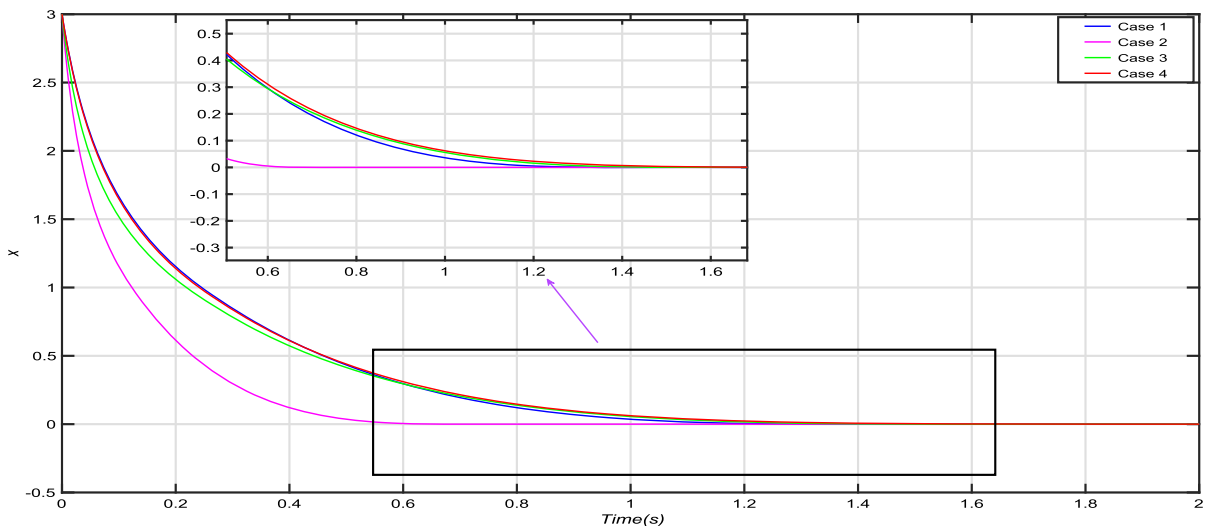


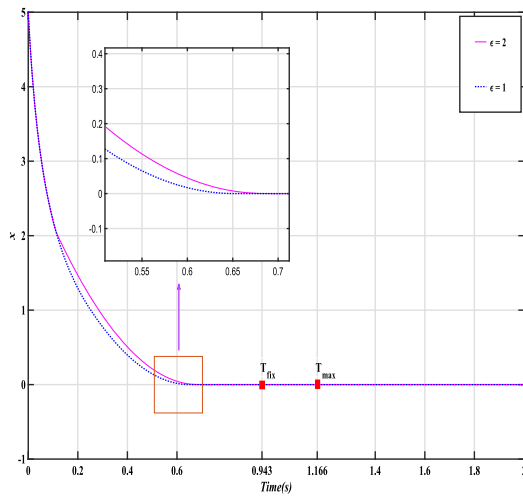
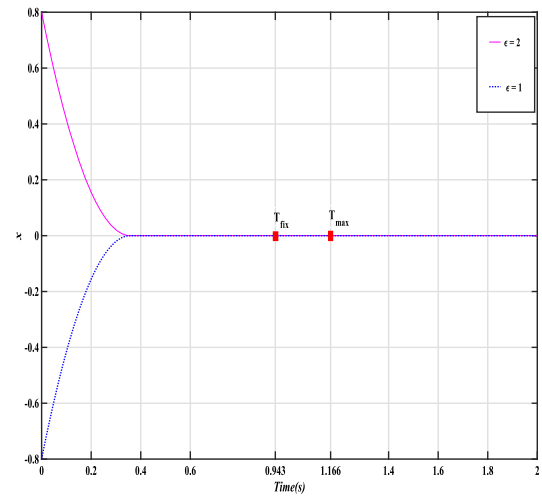
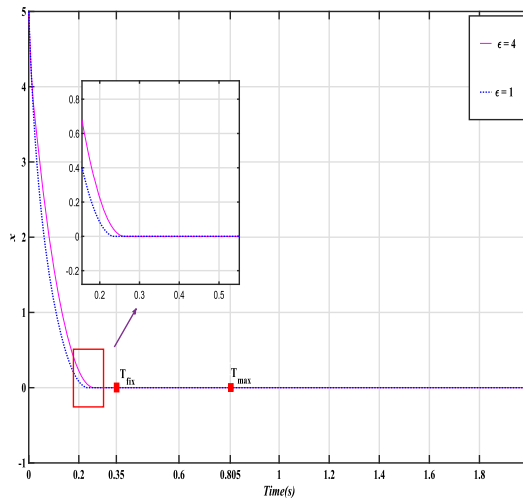
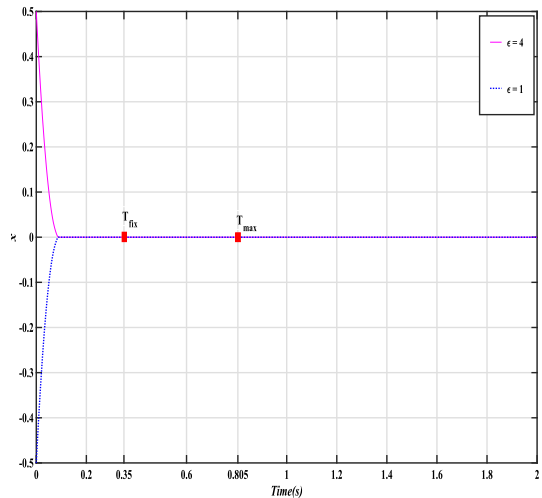
Fig. 5 Convergence rate for different set of sliding surface parameters

Case 2. For, $a < 0$. Let us consider the parameters as $a = -1, b = 14, c = 1, \beta = \frac{3}{2}, \alpha = \frac{1}{2}$. Then by Case 2 of optimality condition we have $\epsilon^{\frac{1}{2}} \left(\epsilon^{\frac{3}{2}} - 1 \right) = 14$ which gives $\epsilon = 4$. So, the upper bound of the settling time as per Theorem 1 is given by $T_{fix} \approx 0.35$, whereas for $\epsilon = 1$ the upper bound is given by $T_{max} \approx 0.805$. Hence we get $T_{fix} \ll T_{max}$. Fig. 7 justifies the claim.

Case 3. For, $a > 0$. Let us consider the parameters as $a = 1, b = \frac{13}{8}, c = 64, \beta = 2, \alpha = \frac{1}{2}$. Then by Case 3 of optimality condition we

have $(64\epsilon^2 - 1)^2 = 64\epsilon \left(\frac{13}{8}\sqrt{\epsilon} - \epsilon \right)$, which gives $\epsilon = \frac{1}{4}$. Therefore, the upper bound of the settling time as per Theorem 1 is given by $T_{fix} \approx 0.752$, whereas for $\epsilon = 1$ the upper bound is given by $T_{max} \approx 1.919$. Hence we get $T_{fix} \ll T_{max}$. Fig. 8 validates this theoretical claim.

Simulation 3: In this section, the effectiveness of the designed sliding surface (30) is shown. In order to verify the effectiveness of the proposed sliding surface, we will be comparing the proposed sliding surface with the two as given in the articles [36] and [37]. We have the

(a) Same absolute values of initial conditions belonging to $(\min\{1, \varepsilon\}, \infty)$ (b) Same absolute values of initial conditions belonging to $(0, \min\{1, \varepsilon\}]$ **Fig. 6** State trajectories of (30) for $\varepsilon = 1$ & $\varepsilon = 2$ corresponding to the parameter $a = 0$ (a) Same absolute values of initial conditions belonging to $(\min\{1, \varepsilon\}, \infty)$ (b) Same absolute values of initial conditions belonging to $(0, \min\{1, \varepsilon\}]$ **Fig. 7** State trajectories of Eq. (30) for $\varepsilon = 1$ & $\varepsilon = 4$ corresponding to the parameter $a = -1$

sliding surface of the present article as S_0 , Eq. (30). The sliding surface presented in article [36] is denoted as S_1 . Then,

$$S_1 = \dot{x} + \theta_1 |x|^{\tau_1} \text{sign}(x) + \theta_2 x + \theta_3 P(x) = 0, \quad (31)$$

where $P(x)$ is auxiliary polynomial function given by:

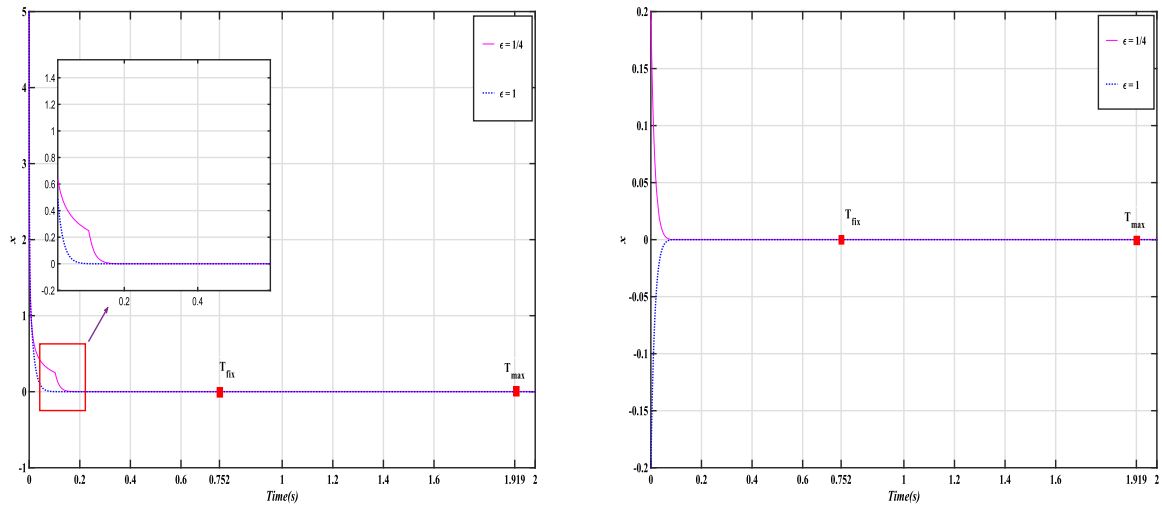
$$P(x) = \begin{cases} |x|^{\tau_2} \text{sign}(x), & |x| \geq \vartheta \\ \rho_1 x + \rho_2 |x|^2 \text{sign}(x) + \rho_3 x^3, & |x| < \vartheta \end{cases},$$

and $\theta_1, \theta_2, \theta_3, \tau_1, \tau_2, \vartheta$ are real constants satisfying $\tau_1 > 1, \theta_1, \theta_2, \theta_3 > 0, \tau_2 \in (0.5, 1), \vartheta \in (0, 1)$. Further, $\rho_1 = \frac{(\tau_2-2)(\tau_2-3)}{2} \vartheta^{\tau_2-1}$, $\rho_2 = -(\tau_2-1)(\tau_2-3) \vartheta^{\tau_2-2}$, $\rho_3 = \frac{(\tau_2-1)(\tau_2-2)}{2} \vartheta^{\tau_2-3}$.

Let the sliding surface in article ([37]) be denoted as S_2 . Then,

$$S_2 = \dot{x} + l_1 |x|^{k_1} \text{sign}(x) + l_2 Q(x) = 0, \quad (32)$$

where $Q(x)$ is a nonlinear function defined as:



(a) Same absolute values of initial conditions belonging to $(\min\{1, \epsilon\}, \infty)$ (b) Same absolute values of initial conditions belonging to $(0, \min\{1, \epsilon\}]$

Fig. 8 State trajectories of Eq. (30) for $\epsilon = 1$ & $\epsilon = 1/4$ corresponding to the parameter $a = 1$

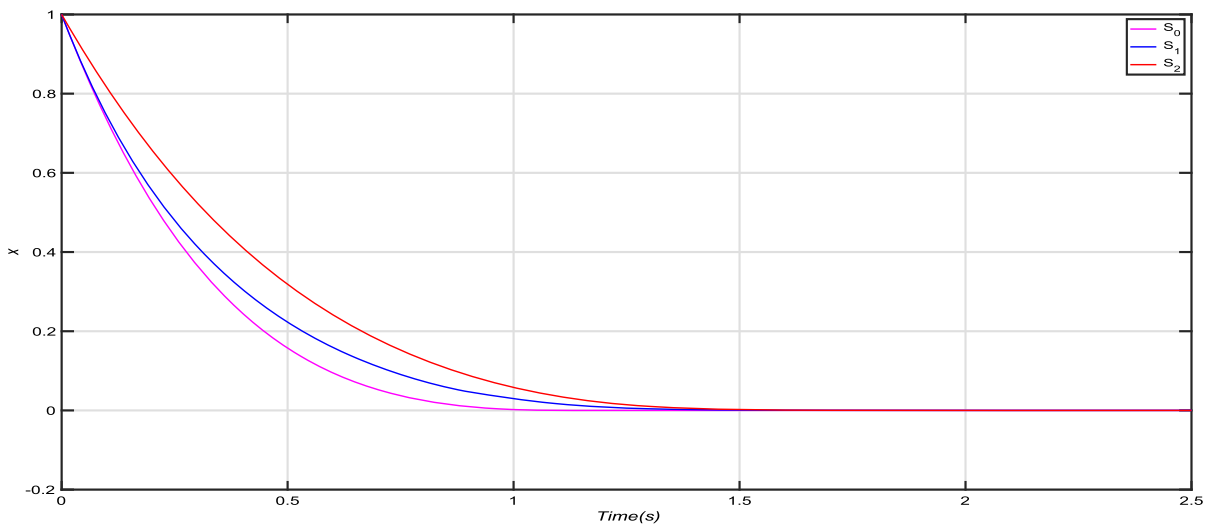


Fig. 9 State trajectories of the sliding surfaces

$$Q(x) = \begin{cases} |x|^{k_1} \operatorname{sign}(x), & |x| > \delta \\ w_1 \sin(x) + w_2 |x|^{m_2} \operatorname{sign}(x), & |x| \leq \delta \end{cases}$$

and $l_1 > 0, l_2 > 0, k_1 = \frac{1}{2}(\psi + 1) + \frac{1}{2}(\psi - 1)\operatorname{sign}(|x| - 1), k_2 = \frac{1}{2}(\psi + \phi) + \frac{1}{2}(\psi - \phi)\operatorname{sign}(|x| - 1), \psi > 1, 0 < \phi < 1$. Further, $0 < \delta < 1, 1 < m_2 < 2, w_1 = \frac{(\phi - m_2)\delta^\phi}{\delta \cos(\delta) - m_2 \sin(\delta)}, w_2 = \frac{\phi\delta^{\phi-1} - w_1 \cos(\delta)}{m_2\delta^{m_2-1}}$.

It should be noted here that for efficient comparison among Eqs. (30), (31) and (32), we should consider $a < 0$. From [36] we have, when the system state stays at a distance from the origin, $\theta_1 |x|^{\tau_1} \operatorname{sign}(x) + \theta_2 x$ dominates and contributes to a high convergence rate. When the system state is close to the equilibrium point, the dominant term $\theta_2 x + \theta_3 P(x)$ guarantees the fixed-time convergence. Therefore, the larger the parameter θ_2 , the faster global convergence rate we will achieve. i.e., the coefficient of the linear term is mostly responsi-

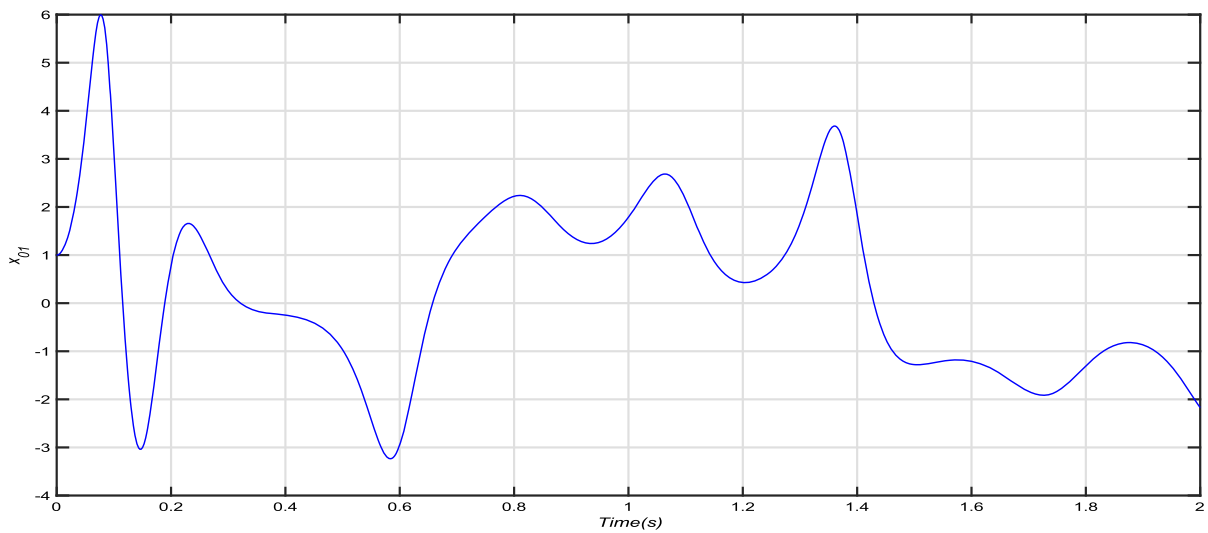


Fig. 10 x_{01} state of the chaotic system (25)

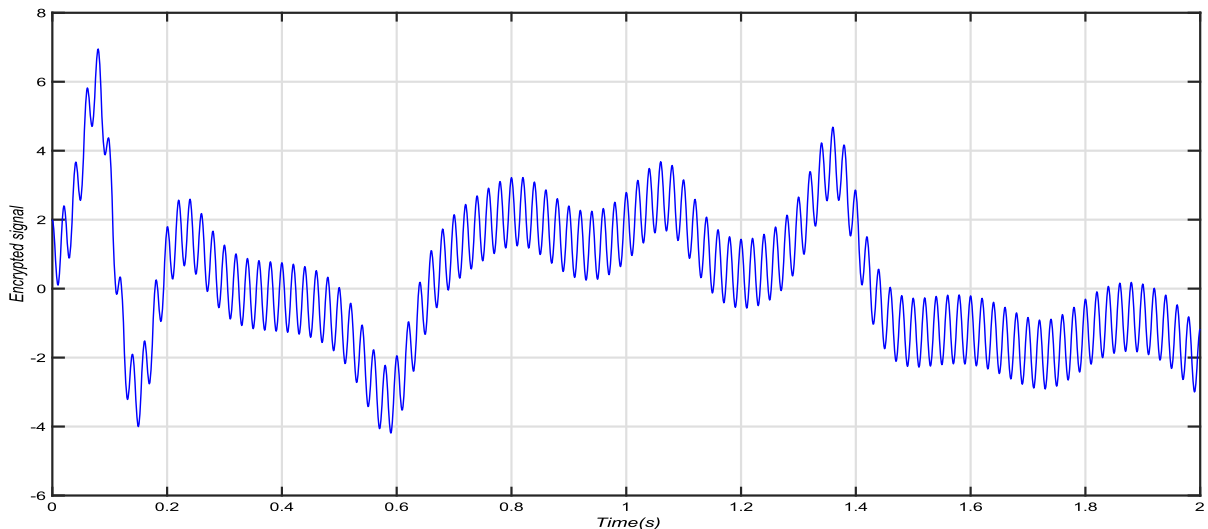


Fig. 11 Masked signal, i.e, message signal + x_{01}

ble for the faster convergence rate. Similarly, from [37] it can be observed that $(l_1 + l_2) |x|^\psi \text{sign}(x)$ dominates to give the fast convergence rate when the system state stays far from origin; whereas $l_2 |x|^\phi \text{sign}(x)$ becomes the dominant term when the states are close to the origin. Hence, greater the value of the parameter l_2 , faster is the global convergence rate. In contrast, from Eq. (30) we have, when the system states are far from origin, $ax + c |x|^{\beta + \text{sign}(|x| - \epsilon)} \text{sign}(x)$ dominates and contributes to a high rate of convergence to reach nearer to the origin. When the system state is close to origin, $ax + b |x|^\alpha \text{sign}(x) + c |x|^{\beta + \text{sign}(|x| - \epsilon)} \text{sign}(x)$ is

responsible for fast fixed-time convergence to origin if $\beta \in (1, 2)$. Hence, the larger the parameter a, c faster is the global convergence rate. From above discussion one can conclude that the global convergence rate can be increased by increasing any of the parameters a, c , for Eq. (30) whereas only one parameter is responsible for the same in Eq. (31) and Eq. (32), thus the proposed sliding surface increases the efficiency of fast global convergence by increasing the liberty to choose one among two parameters.

Also, the state trajectories' strict convergence to the origin corresponding to Eqs. (31) and (32) is not pos-

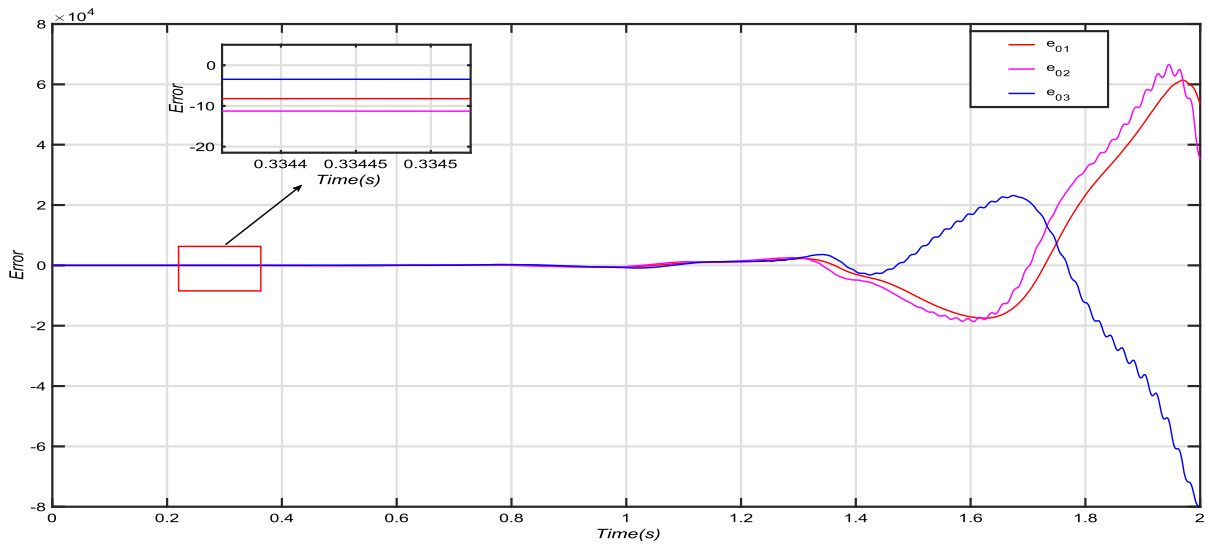
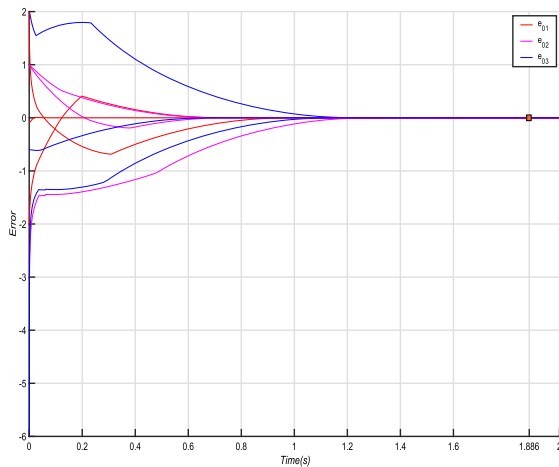
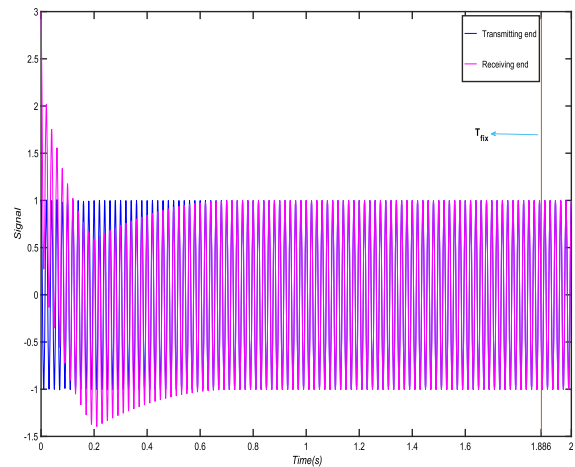


Fig. 12 State trajectories of Eq. (27) without controllers



(a) Stability of state trajectories of Eq. (27)

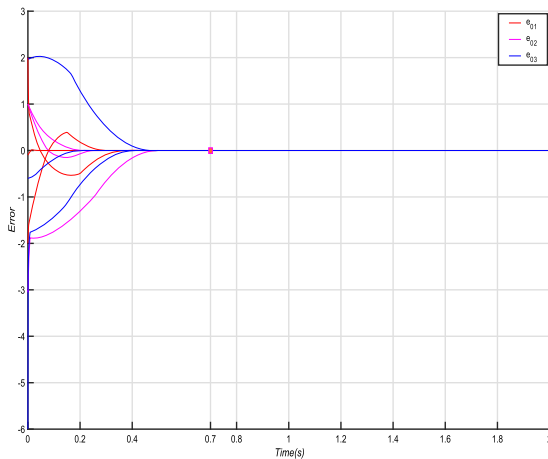


(b) Synchronization of message signal in transmitting and receiving end

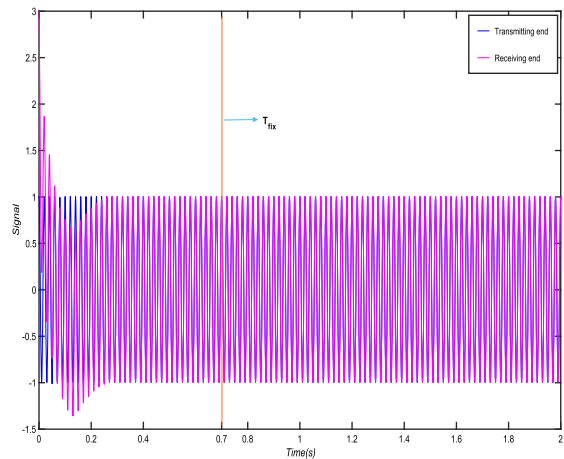
Fig. 13 Stability of trajectories of Eq. (27) and synchronization of transmitted and received signal for $a_0 = 0$

sible; instead, they are bounded in some neighborhood of the origin, whereas the state corresponding to Eq. (30), will converge strictly to the origin in fixed-time. Further, in order to have a fair comparison, the parameters should be selected carefully. To fulfill the mentioned requirement, we will be considering the following set of parameters for Eq. (30), Eq. (31) and Eq. (32). $a = -1$, $b = 1$, $c = 1$, $\beta = \frac{9}{5}$, $\alpha = \frac{5}{9}$, $\epsilon = 1.41$; $\theta_1 = \theta_2 = \theta_3 = 1$, $\tau_1 = \frac{9}{5}$, $\tau_2 = \frac{5}{9}$, $\vartheta = 0.05$, $l_1 = l_2 = 1$, $\phi = \frac{5}{9}$, $\psi = \frac{9}{5}$, $\delta = 0.05$, $m_2 = 1.2$ respectively and the initial condition is taken as $x(0) = 1$. From Fig. 9 one can conclude that though all the trajec-

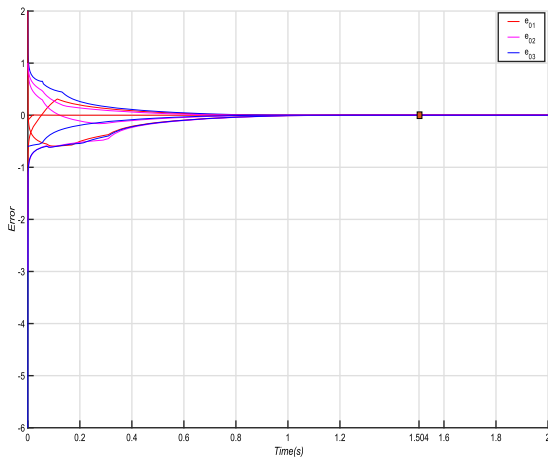
tories corresponding to S_0 , S_1 , S_2 converge too fast, the convergence rate of the trajectory for S_0 is smaller than the other two. Moreover, the upper bound estimates of the settling time corresponding to Eq. (30), Eq. (31) and Eq. (32) can be calculated as $T = 2.04, 2.25$ and 2.18 respectively. Thus, the upper bound estimate of the settling time is more conservative for the Eq. (30). **Simulation 4:** The objective here is to numerically verify the proposition of Sect. 5 in correspondence to the sliding surfaces and controllers designed in Sect. 4. Now, the state trajectory x_{01} of Eq. (25) is depicted by Fig. 10. Whereas, if we add a message signal



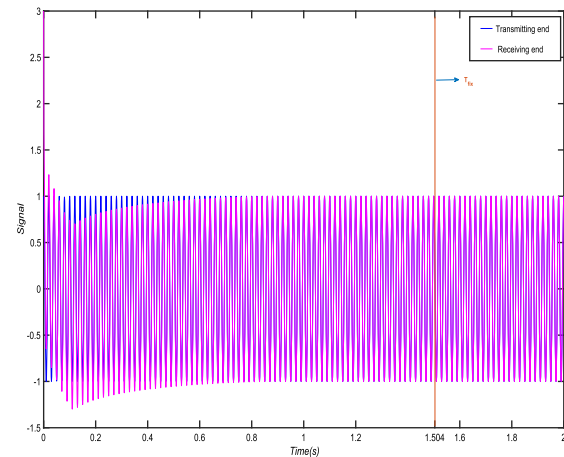
(a) Stability of state trajectories of Eq. (27)



(b) Synchronization of message signal in transmitting and receiving end

Fig. 14 Stability of trajectories of Eq. (27) and synchronization of transmitted and received signal for $a_0 = -1$ 

(a) Stability of state trajectories of (27)



(b) Synchronization of message signal in transmitting and receiving end

Fig. 15 Stability of trajectories of (27) and synchronization of transmitted and received signal for $a_0 = 1$

$m = \cos(100\pi t)$ to the state x_{01} , then the trajectory of $z = x_{01} + \cos(100\pi t)$ is given by Fig. 11. Moreover, as x_{01} is a state of the chaotic system (25) it is very difficult to retrieve the message signal m from the encrypted signal z . Further, we have considered the message signal as $m = \cos(100\pi t)$, the initial conditions of the master and the slave system as $x(0) = (10, -7, 3)^T$ and $y(0) = (1, -5, 1)^T$ respectively. Also the disturbances Δe_{01} , Δe_{02} , Δe_{03} and their bound δ_1 , δ_2 , δ_3 are considered as $1 + 2\sin(t)$, $2 + \cos(t)$, $1 + \cos(t)\sin(t)$ and 4, 4, 3 respectively. For simplicity of calculation we have considered $a_0 = a_1$, $b_0 = b_1$, $c_0 = c_1$, $\alpha = \alpha_1$, $\beta = \beta_0$ and $\epsilon = \epsilon_1$. With this informations, the

trajectories of Eq. (27), without the involvement of controllers, is depicted in Fig. 12. It can be seen from there that the trajectories are unstable and thus are not synchronized when no controllers, given in Sect. 5, is provided in Eq. (27).

Now we will be discussing all the three cases depending on the values of $a_0 = a_1$ to be equal to, less than or greater than zero.

Case 1. $a_0 = a_1 = 0$. Let us consider the parameters as $b_0 = 4$, $c_0 = 1$, $\beta = \frac{3}{2}$, $\alpha = \frac{1}{2}$.

Then by Case 1 of optimality condition we have $\epsilon = 2$. And thus the upper bound is

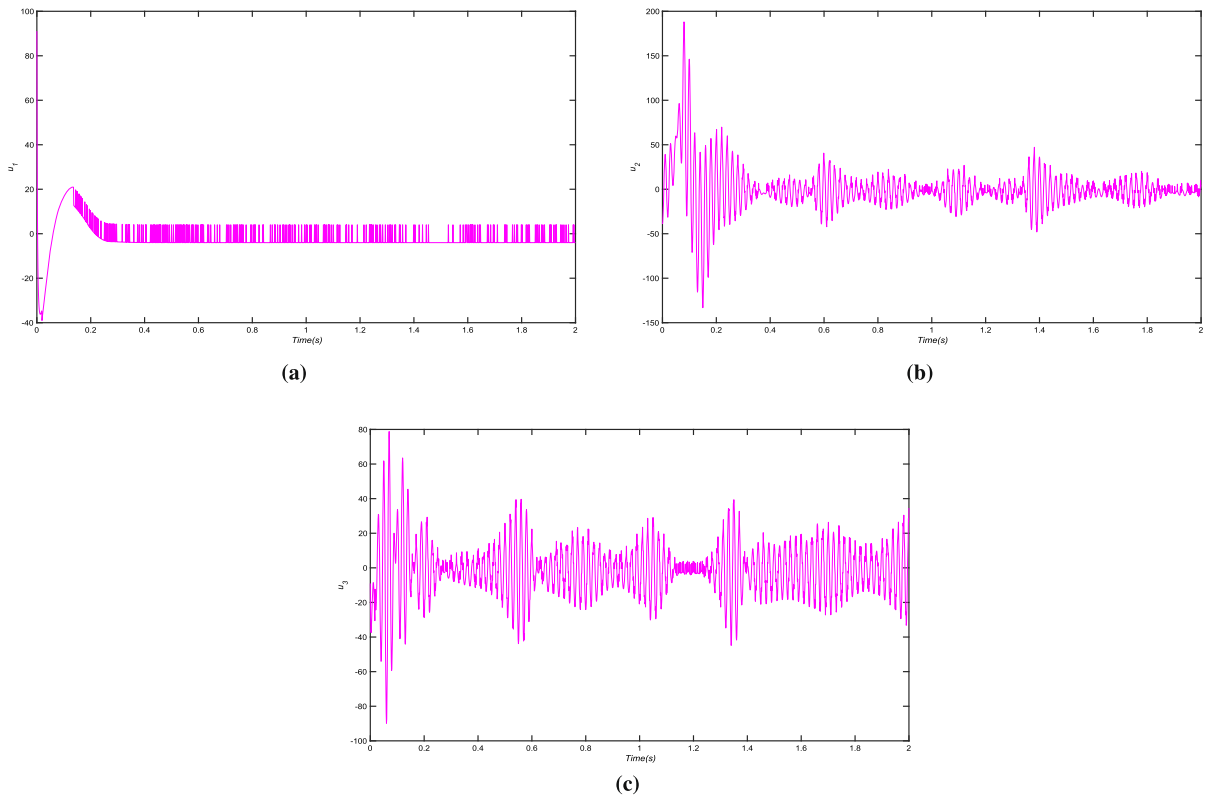


Fig. 16 Control inputs

obtained by $T = T_1 + T_2 = 0.943 + 0.943 = 1.886$.

The results of the synchronization error between the master (25) and slave (26) systems are shown in Fig. 13a. It is visible from there that the synchronization between all the states is adequately achieved within a small time $t < T = 1.886$. Fig. 13b shows that the message signal is retrieved successfully within the time $T = 1.886$.

Case 2. $a_0 = a_1 < 0$. Let us consider the parameters as $a_0 = -1$, $b_0 = 14$, $c_0 = 1$, $\beta = \frac{3}{2}$, $\alpha = \frac{1}{2}$. Then by Case 2 of optimality condition we have $\epsilon^{\frac{1}{2}} \left(\epsilon^{\frac{3}{2}} - 1 \right) = 14$ which gives $\epsilon = 4$. And thus the upper bound is obtained by $T = T_1 + T_2 = 0.350 + 0.350 = 0.7$.

The results of the synchronization error between the master (25) and slave (26) systems are shown in Fig. 14a. From there, it is clear that the synchronization of all the states is successfully accomplished in a brief period of time $t < T = 0.7$. The message signal

m is successfully recovered at the receiver end after a brief interval of time smaller than $T = 0.7$, as seen in Fig. 14b.

Case 3. $a_0 = a_1 > 0$. Let us consider the parameters as $a_0 = 1$, $b_0 = \frac{13}{8}$, $c_0 = 64$, $\beta = 2$, $\alpha = \frac{1}{2}$. Then by Case 3 of optimality condition we have $(64\epsilon^2 - 1)^2 = 64\epsilon \left(\frac{13}{8}\sqrt{\epsilon} - \epsilon \right)$ which gives $\epsilon = \frac{1}{4}$. And thus the upper bound is obtained by $T = T_1 + T_2 = 0.752 + 0.752 = 1.504$.

The results of the synchronization error between the master (25) and slave (26) systems are shown in Fig. 15a. It is visible from there that the synchronization between all the states is achieved properly within the time $t < T = 1.504$. It is apparent from Fig. 15b that the message signal m is successfully recovered at the receiver end within the time $T = 1.504$. The control inputs given in Sect. 5 are depicted through Fig. 16.

7 Conclusion

This study investigates the FXTS of a nonlinear dynamical system and is used to show its application in secure communication systems by utilizing fixed-time non-singular terminal sliding mode control. First, a sufficient criteria is provided on the Lyapunov function to ensure that the state variables can be stable in a set time in accordance with the notion of FXTS, and an upper bound estimate of the settling time is also found. Further, an optimal condition corresponding to a parameter ϵ (keeping others fixed) is derived to obtain the upper bound of the settling time more accurately compared to the article [17], where $\epsilon = 1$ is considered. Second, in correspondence to the provided sufficient condition, a fixed-time NSTSM is derived, which is claimed to give faster convergence as compared to similar sliding surfaces already present in the literature. Also, appropriate sliding mode controllers are designed to force the state variables to the sliding surface within a fixed time. The aforementioned strategy is used in Chaos masking, where an encrypted information signal is decrypted within a certain fixed-time according to the article.

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Data availability The authors declare that all the data generated and analyzed during the study are present in the article.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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